

**Adjective Ordering in Referential Communication:
Subjectivity, Discriminatory Strength, and Incremental Pragmatics**

Hening Wang, Fabian Schlotterbeck, and and Michael Franke

University of Tübingen

Abstract

Adjective ordering preferences are robust across languages, yet speakers also adapt adjective order to the referential context. This paper asks how stable ordering expectations, context-dependent informativeness, and incremental choice jointly shape adjective order in referential communication. We develop a Rational Speech Act model that crosses speaker architecture (global vs. incremental) with semantic regime (context-fixed vs. context-updating), and test it against two German referential experiments: a slider-rating study and a constrained-production study. Across both tasks, order preferences vary systematically with the discriminatory value of the adjectives in the current scene while showing a baseline size-first tendency. Bayesian model comparison shows a clear architecture effect: incremental speakers outperform global speakers in both tasks. By contrast, semantic regime plays a limited role in the fitted models: context-updating and context-fixed semantics are indistinguishable for slider judgments, and context-fixed semantics is slightly favored in production. The production experiment further reveals systematic overspecification, with redundant use of modifiers increasing when size is harder to discriminate; the production model captures this pattern by adding a graded visual-salience layer on top of pragmatic informativeness, allowing perceptual prominence to shape both adjective ordering and selection. Simulation shows that context-updating semantics can nevertheless generate a baseline size-first asymmetry on its own. The results support a multi-level account in which adjective order reflects the interaction of stable ordering expectations, context-sensitive informativeness, perceptual salience, and incremental decision-making. This account connects subjectivity-based explanations of canonical order with efficiency-based accounts of reference while clarifying which mechanisms are needed to predict behavior in judgment and production tasks.

Keywords: adjective ordering, Rational Speech Act, incremental pragmatics, communicative efficiency, subjectivity

Adjective Ordering in Referential Communication: Subjectivity, Discriminatory Strength, and Incremental Pragmatics

1 Introduction

Adjective ordering provides a window onto how speakers coordinate stable linguistic preferences with the demands of situated reference. In a phrase such as *the big blue sticker*, the speaker chooses which properties to mention and the sequence in which those properties will guide the listener’s search. This sequence is constrained by robust ordering preferences: across languages, speakers tend to prefer orders such as “big brown bag” over “brown big bag” (Scontras, 2023; Sproat & Shih, 1991). At the same time, the referential scene creates local pressures on ordering. If colour identifies the target more efficiently than size, the communicatively useful first modifier may conflict with the canonical size-first preference (Fukumura, 2018; Rubio-Fernández, Mollica, & Jara-Ettinger, 2021).

These observations raise a mechanistic question for theories of adjective ordering. Stable semantic or lexical factors account for why some adjective classes tend to occur closer to the noun, while context-sensitive efficiency accounts explain why locally useful properties may be mentioned early. What remains underspecified is how these sources of order preference are combined, and which component of the speaker-listener system carries the context sensitivity. In particular, context sensitivity can enter through semantic composition, when the interpretation of a gradable adjective changes as the comparison class is narrowed, or through speaker choice, when the speaker selects modifiers one at a time in response to the current discriminatory value of each property.

We ask whether context-sensitive adjective ordering is better explained by context-updating semantic composition, by incremental speaker choice, or by both mechanisms together, once stable order expectations and perceptual salience are represented explicitly. If context sensitivity comes from semantic composition, adjective meanings become order-dependent because each modifier is interpreted relative to the comparison class established during composition. If it comes from production, meanings can remain fixed

relative to the scene while the speaker’s word-by-word choices adapt to the listener’s evolving information state.

We address this question in three linked steps. First, the experiments establish which qualitative ordering patterns the model aims to explain: a stable size-first pressure, modulation by referential context, and the richer length and redundancy patterns revealed in production. Second, the model comparison asks which mechanisms improve out-of-sample prediction of these patterns once they are placed in the same RSA architecture. Third, the simulation asks what context-updating composition contributes as a generative mechanism, independently of fitted production parameters.

To implement this comparison, we develop a Rational Speech Act (RSA) model (Degen, 2023; Goodman & Frank, 2016) that crosses two dimensions in a 2×2 design: *speaker architecture* (global vs. incremental) and *semantic regime* (context-fixed vs. context-updating). The global speaker evaluates complete utterances; the incremental speaker chooses one modifier at a time. Under context-updating semantics, the interpretation of an outer adjective such as “big” depends on the comparison class established by the material it composes with, so that “big blue” and “blue big” need not be equivalent; under context-fixed semantics, the size threshold is determined once from the scene. All four model variants are fitted to data from two complementary referential experiments in German — a slider-rating study measuring graded ordering preferences and a constrained production study measuring actual utterance choice — in a shared paradigm that varies discriminatory strength across scenes. Model comparison uses Bayesian leave-one-out cross-validation (PSIS-LOO; Vehtari, Gelman, & Gabry, 2017), which quantifies how much predictive accuracy each mechanism adds. A simulation study applies the same 2×2 crossing outside the fitted empirical models, asking which ordering effects each mechanism can generate on its own and in combination.

Taken together, the experiments, model comparison, and simulation support three conclusions. Across the two experiments, ordering preferences vary systematically with

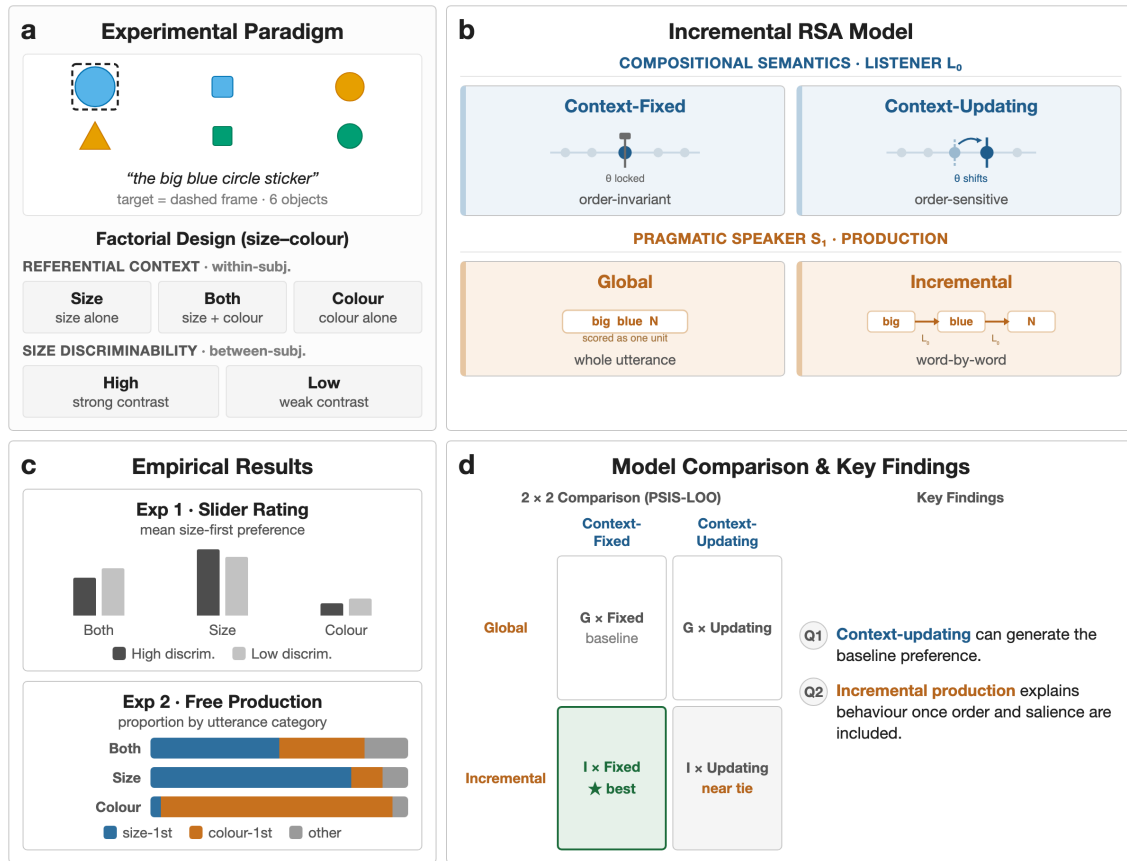


Figure 1

Overview of the study: experimental paradigm, crossed RSA model space, empirical patterns, and 2×2 model comparison.

discriminatory strength: speakers tend to place the more informative adjective earlier, while the canonical size-first preference remains visible as a defeasible baseline. The model comparison shows that predictive fit is driven mainly by speaker architecture: incremental speakers outperform global speakers in both tasks, whereas the semantic-regime contrast is small or absent within the best-fitting incremental models. However, the simulation shows that context-updating composition can still generate a size-first advantage without incremental speaker choice; under context-fixed semantics, a global speaker has no corresponding ordering advantage. Figure 1 provides a schematic overview of the modelling framework, experiments, and key findings.

The remainder of the paper is structured as follows. Section 2 reviews prior work on adjective ordering, referential efficiency, and incremental pragmatic modelling. Section 3 introduces the modelling framework, and Section 5 gives the formal specification. Section 4 reports the two referential studies. Sections 6 and 7 present the fitted model comparison and simulation study. Section 8 summarizes the results and discusses their implications.

2 Background

Research on adjective ordering has identified several stable predictors of preferred order, including meaning-based factors such as closeness to the noun and inherentness (Martin, 1969; Sweet, 1898; Whorf, 1945), class-based hierarchies (Cinque, 1994; Dixon, 1982; Quirk, Greenbaum, Leech, & Svartvik, 1972; Scott, 2002), and information-theoretic measures (Dyer, Torres, Scontras, & Futrell, 2023; Futrell, Dyer, & Scontras, 2020; Hahn, Degen, Goodman, Jurafsky, & Futrell, 2018). These predictors explain substantial structure, but residual variation remains even for strong individual predictors (for an overview, see Dyer et al., 2023; Scontras, 2023). Ordering preferences are also context-sensitive in ways that fixed hierarchies do not fully capture (Fukumura, 2018; Trotzke & Wittenberg, 2019). The present account therefore draws on three lines of work that speak directly to the source of this context sensitivity: subjectivity-based composition, referential efficiency, and incremental pragmatic choice.

2.1 Subjectivity and Adjective Ordering

A central empirical generalization in the adjective-ordering literature is the relation between an adjective’s *subjectivity* and its preferred distance from the noun. Scontras, Degen, and Goodman (2017) operationalized subjectivity as susceptibility to faultless disagreement and showed that it explains a large share of variance in English ordering preferences, outperforming earlier predictors such as inherentness and intersectivity. Subsequent cross-linguistic work suggests that this relationship is not peculiar to English and may reflect a recurring regularity in adjective ordering (Scontras, 2023; Trainin & Shetreet, 2021). This relation is directly relevant to size–colour sequences because size adjectives are typically

more subjective and gradable than colour adjectives, yielding an expected size-first baseline in prenominal modification.

Scontras, Degen, and Goodman (2019) argue that the explanatory core of this effect lies in restrictive composition. Assuming a hierarchical prenominal structure, the adjective closest to the noun composes first with the nominal denotation, and the outer adjective applies to the result of that restriction. The outer, more subjective adjective is therefore evaluated against a comparison class that has already been narrowed by the inner, less subjective adjective. In such a restricted comparison class, placing the less subjective adjective closer to the noun is advantageous because its interpretation provides a more stable basis for subsequent gradable modification. Franke, Scontras, and Simonic (2019) make the same intuition explicit in simulation, showing that subjectivity-aligned orders increase average communicative success under sequential composition with noisy meanings. For the present question, this explanation locates the ordering pressure in the compositional system rather than in a word-by-word production policy.

Scontras, Bar-Sever, Kachakeche, Rosales Jr, and Samonte (2020) develop this idea as *incremental semantic restriction*. If nominal structure supports sequential narrowing of the comparison class, then subjectivity-based ordering pressures should arise even without an incremental speaker in the usual process-model sense. This identifies a plausible compositional source for canonical ordering in the present paradigm: context-sensitive restrictive composition can favor placing the more subjective, gradable size adjective outside the less subjective colour adjective.

2.2 Communicative Efficiency and Discriminatory Strength

A second source of ordering pressure comes from referential efficiency. A separate line of work has shown that adjective use is sensitive to the demands of the immediate communicative context, including whether a modifier helps the listener identify the referent efficiently (Degen, Hawkins, Graf, Kreiss, & Goodman, 2020; Rubio-Fernández, 2016; Rubio-Fernandez, 2019). This literature treats adjective choice and adjective order as part of

a broader problem of reference generation rather than as the expression of a static lexical hierarchy; a related commitment appears in Van Gompel and colleagues' conceptualization model, which derives property choice from the contrast set available to the speaker (Fukumura, Van Gompel, & Pickering, 2010; Van Gompel, Van Deemter, Gatt, Snoeren, & Krahmer, 2019).

Fukumura (2018) provide a direct empirical precedent for the present experiments. In their production studies, speakers were more likely to place the more discriminating adjective first, consistent with a *discriminatory efficiency hypothesis*: order should reflect which property best distinguishes the target in the current scene. The relevant pragmatic weight is therefore computed from the contrast set on the current trial. The same adjective may be highly useful in one array and much less useful in another.

Rubio-Fernández and colleagues develop this idea within a broader account of *incremental efficiency* (Rubio-Fernandez & Jara-Ettinger, 2020; Rubio-Fernández et al., 2021). They argue that the efficiency of a referential expression should be evaluated as the utterance unfolds, before the full description has been heard. This production-side view is closely aligned with incremental interpretation work showing that listeners use visual context and pronominal modifiers to restrict the candidate set before the noun is complete (Sedivy, Tanenhaus, Chambers, & Carlson, 1999; Tanenhaus, Spivey-Knowlton, Eberhard, & Sedivy, 1995). From this perspective, word order matters because it determines how the listener's search space is narrowed over time. Prenominal and postnominal systems therefore create different pragmatic affordances, and even redundant modifiers can be useful if they facilitate early search. The relevant quantity is the discriminatory value a property has at the stage of processing where it is encountered.

This work motivates treating discriminatory strength as a source of graded context sensitivity. The relevant test is whether size-first preferences persist, weaken, or reverse as size and colour differ in their usefulness for identifying the target, including when the size contrast itself is easier or harder to discriminate.

2.3 Incremental Pragmatic Models

The Rational Speech Act (RSA) framework provides the formal basis for comparing these pressures. In RSA, a pragmatic speaker selects utterances by reasoning about a listener’s posterior, and the listener in turn reasons about the speaker’s choice (Degen, 2023; Goodman & Frank, 2016). Although standard RSA treats complete utterances as the units of choice, the same logic can be defined incrementally, with the speaker choosing one word at a time on the basis of how each token updates the listener’s beliefs (CohnGordon, Goodman, & Potts, 2019). Existing incremental pragmatic models and related work on redundancy show that word-by-word reasoning can capture asymmetries between adjective positions and between prenominal and postnominal systems (Degen et al., 2020; Rubio-Fernández et al., 2021). These accounts typically update the listener’s posterior online while keeping adjective meanings fixed relative to the scene. Schlotterbeck and Wang (2023) introduced a closely related alternative, together with the slider-rating paradigm that forms the basis for one of the experiments reported here. In that model, the threshold for a gradable adjective is recomputed from the listener’s current posterior at each stage of composition. The study provided a qualitative fit in which the model captured both a subjectivity-driven baseline ordering and a discriminatory-strength modulation. It did not separate context-updating semantic composition from incremental speaker choice in a joint quantitative comparison, so the same qualitative pattern remained compatible with multiple mechanisms.

3 The Modelling Framework

We extend Schlotterbeck and Wang (2023) by separating the mechanisms left open by the preceding literature. The framework embeds their proposal in a systematic 2×2 comparison of speaker architecture and semantic regime, including a context-fixed ablation. It adds a constrained production experiment with a richer response space and a simulation study that asks which ordering effects follow from context-updating composition outside the fitted empirical models. Bayesian model comparison then quantifies which mechanisms add predictive accuracy once production-specific order and salience pressures are represented

explicitly. Here we describe the framework at a conceptual level; the meaning functions, speaker definitions, and inference setup are given formally in Section 5. The remaining subsections develop these components in turn.

3.1 Reference, Scenes, and the Speaker’s Problem

The model addresses a reference game. A scene contains several objects that vary in size, colour, and form, and one of them is the intended referent — in our materials, the largest object that matches the target colour and form. A speaker must choose an adjective string that lets a listener identify this referent, and the listener interprets that string by progressively narrowing the set of objects it could pick out. Because objects differ along three properties, the same referent can be described in many ways, and the order in which adjectives are produced is itself part of what the speaker chooses.

3.2 Two Mechanisms for Order Sensitivity

Determinate properties such as colour and form are treated as nearly categorical, whereas size is gradable. Whether an object counts as “big” depends on the comparison class it is evaluated against, so the size term is informative only relative to the objects currently under consideration. Against this background, the framework isolates two ways in which adjective order can come to matter. The first is the *speaker architecture*. A global speaker evaluates complete adjective strings and chooses among them as wholes, whereas an incremental speaker chooses one adjective at a time, weighing how the next word would reshape the listener’s beliefs at each step. For an incremental speaker, a modifier is evaluated at the moment it is produced, according to the belief update it would cause before later words are heard. The second is the *semantic regime*. Under context-fixed semantics, the size threshold is computed once from the scene and held constant, so reordering the adjectives cannot change what “big” means. Under context-updating semantics, the threshold is recomputed from the listener’s running beliefs. Once “blue” has restricted attention to blue objects, “big” is judged against just those objects; “big blue” and “blue big” can then differ even though they contain the same words. This non-commutativity connects to Teodorescu’s

(2006) observation that ordering restrictions weaken precisely when the alternative orders are not truth-conditionally equivalent.

Crossing these two binary distinctions yields a 2×2 space of models — speaker architecture (global vs. incremental) by semantic regime (context-fixed vs. context-updating). The global context-fixed model is the baseline and the closest to the vanilla RSA model: meanings are fixed and the speaker evaluates complete strings, so any order preference comes from the shared order prior. The global context-updating model adds posterior-dependent meanings, the incremental context-fixed model adds word-by-word speaker choice, and the incremental context-updating model combines both sources. All four models include the same basic ingredients needed to connect the model to the experimental task: a prior over adjective orders and a utility term based on referential informativeness. The production models additionally include visual salience and stopping components because the production task allows one-, two-, and three-adjective strings; these components are held constant across models so that the 2×2 comparison isolates the contribution of semantic regime and speaker architecture.

3.3 Qualitative Predictions

This framework yields qualitative predictions that the two experiments are designed to test (Table 1). A stable, subjectivity-aligned ordering prior yields a baseline size-first preference that survives even when colour alone identifies the referent (P1). The pragmatic component is sensitive to which property discriminates the referent in the current scene, so the size-first preference should weaken as colour becomes the more useful property (P2). The size-discriminability manipulation provides a proxy-based diagnostic extension of this logic: if lower visual reliability behaves like a display-level analogue of subjectivity, then low-discriminability size contrasts should strengthen the size-first preference (P3). This prediction extrapolates from the compositional logic in Franke et al. (2019); Scontras et al. (2020): if lower discriminability makes size interpretation noisier, placing colour closer to the noun restricts the comparison class before size is evaluated, which corresponds to a size-first

order preference in the German prenominal sequence. Finally, because production allows the full inventory of one-, two-, and three-adjective strings, the framework predicts a graded over-informativity gradient that the binary slider task cannot reveal; if low size discriminability makes size less reliable for reference, low-discriminability displays should elicit longer, more redundant strings (P4).

Table 1

Qualitative predictions of the modelling framework and where each is tested.

Label	Prediction	Tested in
P1	A baseline size-first pressure is detectable even when colour alone identifies the referent.	Exp. 1 & 2
P2	The size-first preference weakens as colour becomes the more discriminating property.	Exp. 1 & 2
P3	If low visual size discriminability behaves like a display-level proxy for subjectivity, it yields a stronger size-first preference.	Exp. 1 & 2
P4	The production task exposes a graded over-informativity gradient across one-, two-, and three-adjective strings, and that gradient is modulated by size discriminability.	Exp. 2

4 Experiments

We tested the qualitative predictions of the modelling framework (Table 1) with two complementary online referential tasks in German. Experiment 1 measured explicit order preferences in a slider-rating task, whereas Experiment 2 measured adjective selection and ordering in constrained production. Comparing the two tasks lets us ask whether the same context manipulations affect both direct judgments between competing orders and speakers' own adjective choices.

4.1 Shared Method

4.1.1 *Design*

Both experiments shared the same basic referential design. The response format differed across experiments, but the visual displays and target-definition logic were shared. On every critical trial, participants saw a six-object display with one framed target. The target was always the largest object that matched the intended colour and form. The analyses reported here focus on the theoretically central size–colour subset.¹ Within that subset, referential context varied within participants and had three levels: size-sufficient, colour-sufficient, and both-necessary. A second, between-participant factor, size discriminability, varied the size gap between the target and its size competitors. High-discriminability displays used larger target–competitor gaps, making the size contrast visually pronounced, whereas low-discriminability displays used smaller gaps, making size a less reliable cue. This factor was manipulated between participants so that each participant experienced a consistent level of visual size contrast throughout the task (Table 2).

Table 2

Shared design of the two experiments.

Factor	Levels
Referential context	Size-sufficient, colour-sufficient, both-necessary
Size discriminability	High (strong size contrast), low (weak size contrast)

¹ The broader paradigm also included size–form and colour–form trials to widen the material base and reduce dependence on a single adjective pairing. These conditions remain part of the experimental design, but the present analysis concentrates on the size–colour subset because it provides the clearest contrast between a gradable adjective and a determinate feature adjective. The exact size values, target–competitor gap construction, and list assignment are reported in Appendix A.

4.1.2 Materials

An example item is shown in Figure 2. Items were distributed across participants in a Latin-square design, so that scene templates and condition assignments were counterbalanced across lists. In addition to list counterbalancing, the critical scenes were realised through systematic permutations of colour and form values, creating variability in the surface displays while preserving the referential structure required by each condition. These permutations were applied to 27 critical scene templates, each specifying an abstract six-object configuration with a target, competitors, and distractors. Critical vocabularies were fixed across the study, with six colour terms and six form terms reused throughout the scene set; the modified noun was always *Aufkleber* (‘sticker’).² For the computational model, each display was represented as a 6×3 matrix, with one row per object and three columns for size (*s*), colour (*c*), and form (*f*). The referent was stored at index 0, but participants saw the displays as ordinary visual arrays rather than as explicit feature lists.

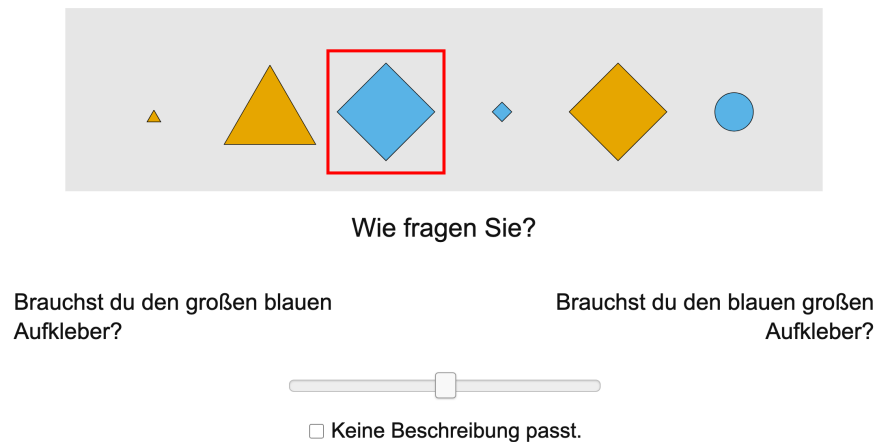


Figure 2

Example slider-rating trial from the experiment.

² The form values were presented with German adjectival forms, such as *runden* for round/circle and *quadratischen* for square, so that form could occupy the same prenominal adjective position as colour and size.

4.1.3 Procedure

Participants completed the experiments online via Prolific and were instructed to identify the framed target using its perceptual properties, not spatial descriptions such as *left* or *right*. The task was designed as a simple reference game: participants either evaluated or produced a description that would allow another person to identify the target accurately and efficiently. The cover story framed each trial as a telephone conversation about collectible stickers: participants saw a set of six stickers on screen, one highlighted by a coloured frame, and were asked to refer to the target for a partner who could see the same stickers in a different order but could not see the frame. Both experiments began with a consent form and a brief practice block (three training trials with feedback) that familiarised participants with the interface and the referential task. The main trials were divided into four blocks of approximately equal length, with self-paced breaks between blocks. Trial order within each block was randomised independently for each participant. Experiment 1 presented two competing adjective orders on a continuous slider and asked participants which description would better identify the target for their partner. It contained 180 trials in total: 81 critical trials and 99 filler or control trials, drawn from one of six counterbalanced lists. Experiment 2 used the same list structure for the critical trials but asked participants to type the adjective material they would place before the noun *Aufkleber* to describe the target to their partner. Each participant completed 135 main trials, including 81 critical trials and 54 filler or control trials.³

4.2 Experiment 1: Slider-Rating Study

4.2.1 Participants

Participants ($N = 134$ after exclusions; 70 low discriminability, 64 high discriminability) were recruited online via Prolific and reported German as their native or fluent language. Exclusion criteria were: more than three missing trial-list entries,

³ The number of filler and control trials was reduced in Experiment 2 because typed production responses were slower and more effortful than slider judgments.

completion times more than two standard deviations from the sample mean, more than three failures on control items, or more than three “none fits” responses on experimental items. The analysed dataset contained 10,606 critical observations. The size–colour subset reported in the main empirical and model-comparison figures contained 3,485 observations; exclusion details are summarized in Appendix A. Mean completion time was 27.6 minutes. Participants were compensated via Prolific at a rate above the platform-recommended minimum.

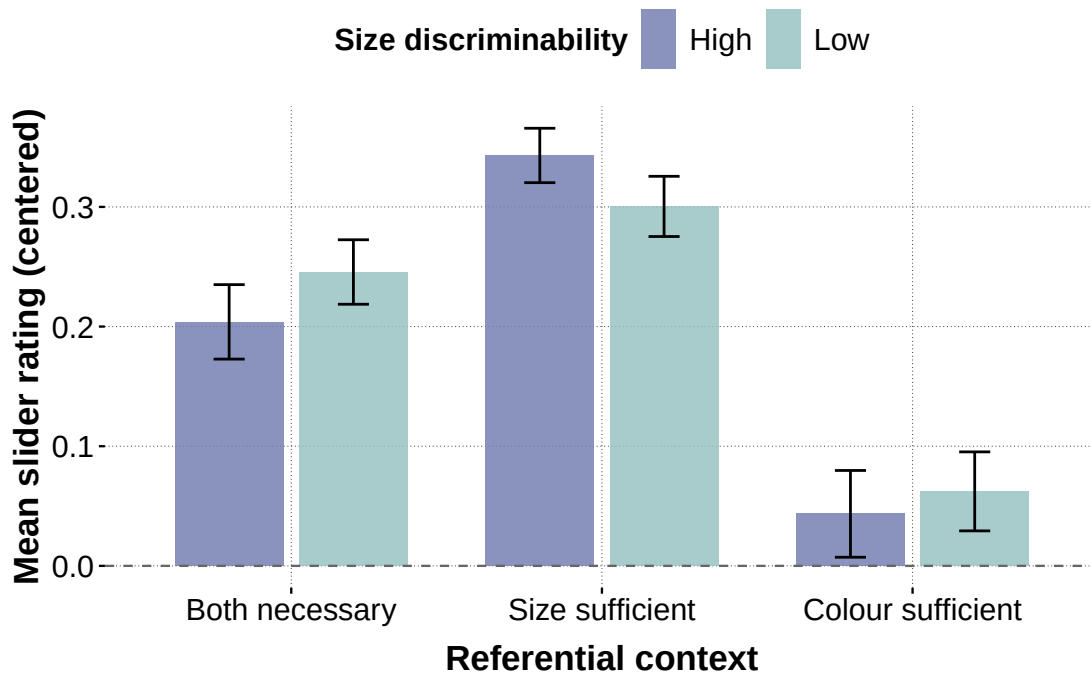
4.2.2 Procedure

Experiment 1 implemented the shared referential design as a comparative judgment task. Participants were asked to imagine a referential exchange about sticker collections and to evaluate which description would be better for identifying the framed target. On each critical trial, two competing adjective orders appeared on opposite sides of a continuous slider, together with a “none of the descriptions fits” option. Left–right presentation of the two orders was randomised across trials. Each participant saw one of six lists built from the 27 underlying scenes, yielding 81 critical trials and 99 filler and control trials (180 total). The filler and control items discouraged fixed response strategies and ensured attention to the referential task. Raw slider values were recorded on a 0–100 scale and rescaled to $[-0.5, 0.5]$, with positive values indicating a size-first preference. For the ZOIB model, the same responses were mapped to $[0, 1]$, so the midpoint/no-preference point is 0.5 in the posterior-predictive figures. The descriptive analysis used a linear mixed-effects model with centred slider rating as the dependent variable and referential context, size discriminability, and their interaction as fixed effects, with random intercepts for participant and item. These mixed-effects regressions provide compact descriptive summaries; the main mechanism-level claims are evaluated by the Bayesian model comparisons reported in Section 6. For the Bayesian model comparison, these ratings were modelled with a Zero-One-Inflated Beta (ZOIB) likelihood to accommodate boundary mass at 0 and 1.

4.2.3 Results

Figure 3 shows that size-first orderings were preferred across all three referential contexts, but the strength of the preference depended on which property was informative. Referential context produced a large main effect, $F(2, 3349) = 213.7$, $p < .001$; $\chi^2(2) = 402.6$, $p < .001$. The size-first preference was strongest in size-sufficient displays, intermediate in both-necessary displays, and weakest in colour-sufficient displays. Notably, even colour-sufficient contexts did not produce a qualitative reversal: mean centred ratings remained above zero.

Size discriminability did not produce a significant main effect, $F(1, 132) = 0.03$, $p = .86$, but it interacted with referential context, $F(2, 3349) = 5.84$, $p = .003$; $\chi^2(2) = 11.7$, $p = .003$. The interaction was asymmetric: higher discriminability slightly increased the size-first preference in size-sufficient contexts (low: 0.30, high: 0.34) but slightly reduced it in both-necessary (0.25 vs. 0.20) and colour-sufficient contexts (0.06 vs. 0.04). Post hoc pairwise comparisons (Bonferroni-corrected) showed that no individual simple effect of discriminability within any referential context reached significance (all $p > .12$), indicating that the interaction is driven by the directional pattern across contexts rather than by any single condition contrast. This pattern is consistent with a discriminatory-strength interpretation: ordering preferences track the usefulness of size for identifying the target, but a baseline size-first tendency remains detectable even when colour is locally more discriminating. P1 and P2 are supported in the slider data; the display-level proxy prediction for P3 is not. Size discriminability produces no main effect, no simple effect within any referential context reaches significance, and the asymmetric interaction reverses direction in size-sufficient displays, where higher discriminability slightly strengthens rather than weakens size-first ratings. The interaction indicates a small context-linked modulation of size-first ratings, with limited support for a monotonic subjectivity-aligned effect.

**Figure 3**

Slider-rating data. Mean centred slider ratings by referential context and size discriminability, with 95% confidence intervals; positive values indicate size-first preference.

4.3 Experiment 2: Constrained-Production Study

4.3.1 Participants

Participants ($N = 113$ after exclusions; 60 low discriminability, 53 high discriminability) were recruited online via Prolific; anyone who had completed the slider study was excluded. Further exclusion criteria were: completion times more than two standard deviations from the mean, more than 80% identical responses, or more than 10% missing or unannotatable responses. The broad preprocessed critical table contained 9,639 rows across the three adjective-pair families. The size-colour production model reported in Section 6.2 used 3,196 annotated observations; exclusion and coding details are summarized in Appendix A. Mean completion time was 32.4 minutes. Compensation followed the same Prolific rate as Experiment 1.

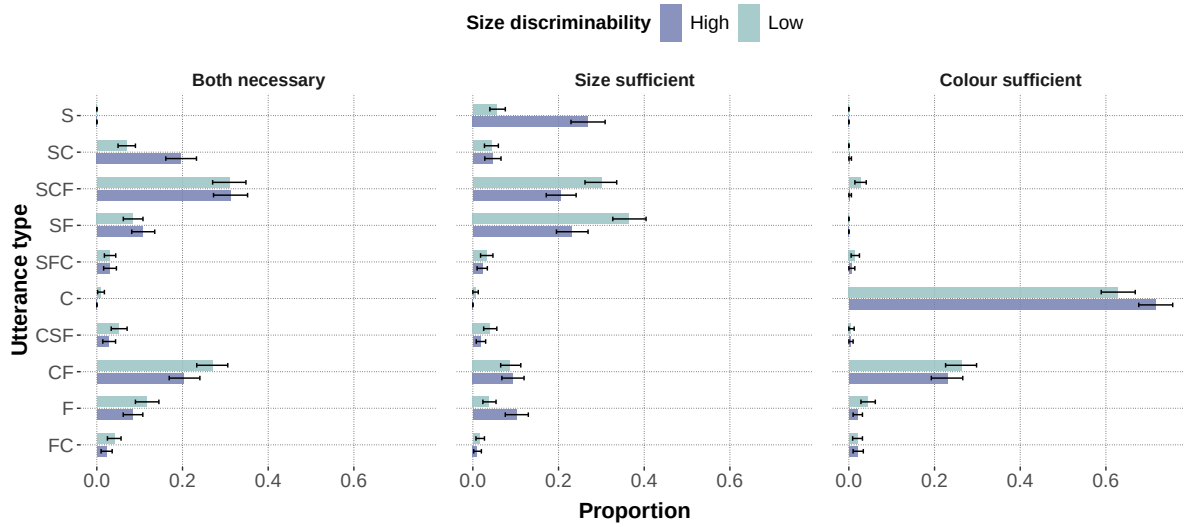
4.3.2 Procedure

Experiment 2 used the shared referential design in a constrained production format. Participants saw the framed target together with a sentence prompt and typed the adjective material needed before the noun *Aufkleber*. Restricting the response to the adjective slot rather than eliciting a full sentence reduced off-task variability and provided a controlled basis for coding. Instructions again prohibited spatial descriptions, and a practice block familiarised participants with the relevant colour and form vocabulary before the critical trials began. Each list contained 81 critical trials; the filler load was reduced relative to Experiment 1 (135 total trials vs. 180) to accommodate the greater time burden of typed production.

Responses were coded into 15 utterance categories corresponding to all one-, two-, and three-adjective strings built from size, colour, and form terms (3 one-adjective, 6 two-adjective, 6 three-adjective). Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous; spatial descriptions, comparative or ordinal size expressions, and other unclassifiable responses were excluded. The dependent variable for the Bayesian model comparison is the categorical utterance label $y_i \in \{1, \dots, 15\}$ for each trial i . For descriptive statistics and targeted mixed-effects analyses, we also tracked two derived binary outcomes: whether an utterance was size-initial and whether it was over-informative (i.e., a three-adjective response in both-necessary contexts, or a multi-adjective response in size-sufficient or colour-sufficient contexts).

4.3.3 Results

Production showed a more categorical context pattern than the slider ratings. As Figure 4 illustrates, size-first utterances dominated in size-sufficient contexts, colour-first utterances dominated in colour-sufficient contexts, and both-necessary contexts fell between the two extremes. A mixed-effects logistic regression on size-initial responses yielded a strong main effect of referential context, $\chi^2(2) = 1859.7$, $p < .001$, no significant main effect of size discriminability, $\chi^2(1) = 0.79$, $p = .37$, and a significant interaction, $\chi^2(2) = 31.4$, $p < .001$.

**Figure 4**

Constrained-production data. Utterance-type proportions by referential context and size discriminability, with bootstrapped 95% confidence intervals; only utterance types reaching at least 2% in one condition are shown.

The interaction was again asymmetric, although its cellwise direction differed from the slider task: higher discriminability reduced the size-first advantage in size-sufficient contexts but increased it in both-necessary contexts.

A complementary analysis of over-informativity reveals a related efficiency pressure. Over-informative utterances were most frequent in size-sufficient contexts and least frequent in colour-sufficient contexts, producing a strong main effect of referential context, $\chi^2(2) = 725.7$, $p < .001$. Unlike size-initiality, over-informativity also showed a main effect of size discriminability, $\chi^2(1) = 10.3$, $p = .001$, with more multi-adjective responses in low- than in high-discriminability displays. This effect interacted with referential context, $\chi^2(2) = 63.8$, $p < .001$, driven primarily by size-sufficient contexts. The pattern aligns with the broader communicative-efficiency literature on redundancy: speakers produce additional material when it helps resolve reference, even if it is not strictly necessary for uniqueness. The production data support P2 strongly. They also show why P1 should be treated as a baseline pressure rather than as a categorical prediction: size-first utterances dominate when size is

useful, but colour-first utterances dominate when colour alone identifies the referent. The display-level proxy prediction for P3 is again not supported. The interaction on size-initial responses is asymmetric, with higher discriminability reducing size-first only in size-sufficient contexts and increasing it in both-necessary contexts, so lower visual size reliability does not monotonically promote size-first ordering. P4 is supported: the richer response space reveals a graded over-informativity gradient across one-, two-, and three-adjective strings, and low-discriminability displays elicit more redundant strings overall.

4.4 Interim Summary

Across both experiments, three empirical regularities emerge. First, there is a baseline size-first pressure that is clearest in the slider judgments and is overridden in production when colour is the only discriminating property. Second, ordering is modulated by referential context: size-first responses increase when size is informative and decrease when colour is more useful, consistent with a discriminatory-efficiency account. Third, size discriminability does not produce significant main effects, and the direction of its interaction with referential context varies across contexts: in the slider task, higher discriminability raised size-first ratings in size-sufficient displays but lowered them in both-necessary and colour-sufficient displays, while in production the directions were reversed for size-sufficient and both-necessary displays. The data therefore weigh against the proxy prediction that lower visual size reliability yields a uniformly stronger size-first preference. The production data sharpen the first two patterns relative to the slider data: typed production yields more categorical context effects and additionally reveals an over-informativity gradient that is sensitive to size discriminability, with low-discriminability displays eliciting longer, more redundant strings. The failure of this proxy prediction should be interpreted narrowly. The high- versus low-discriminability manipulation changes the visual reliability of size in a scene; it is not a direct manipulation of lexical subjectivity itself. Thus the data weigh against a monotonic effect of visual size reliability on ordering, while leaving intact the broader claim that subjectivity-aligned expectations contribute to the stable size-first baseline. Together,

these results motivate a formal model that can capture both the stable ordering bias and its context-dependent modulation, while accommodating the richer response space of constrained production.

5 Computational Model

This section turns the modelling framework introduced in Section 3 into a formal RSA model. The formalization separates components shared across model variants from components that instantiate the two tested contrasts: speaker architecture and semantic regime. The section specifies the representational primitives, literal-listener semantics, global and incremental speakers, context-fixed and context-updating regimes, and inference setup for the 2×2 model comparison. The experiments were broader than the fitted comparison: the full paradigm included size–form and colour–form trials, but the analyses reported here focus on size–colour scenes, where the contrast between a gradable adjective and a determinate feature adjective is clearest. Form remains part of the representational vocabulary and production response space, but it is not the primary mechanistic contrast in the current fit.

5.1 Representational Primitives

An object is a triple $o = (s, c, f)$, where $s \in [1, 10]$ denotes size, $c \in \{0, 1\}$ indicates colour match, and $f \in \{0, 1\}$ indicates form match. A context $\mathcal{C}$ is a set of n_{obj} objects. In the experiments, the intended referent is always the largest object with $c = f = 1$. The size–colour fits use C as the determinate feature that competes with size for identification. The formal vocabulary still includes F because form was available in the broader experiment and could appear as an additional, potentially redundant modifier in production.

Utterances are built from three adjective types, S, C, and F. An utterance $u = (a_1, \dots, a_T)$ is an ordered sequence of adjectives $a_t \in \{\text{S}, \text{C}, \text{F}\}$ with $T \in \{1, 2, 3\}$. Combining all lengths and orderings yields 15 surface forms: $\{\text{S}, \text{SC}, \text{SCF}, \text{SF}, \text{SFC}, \text{C}, \text{CS}, \text{CSF}, \text{CF}, \text{CFS}, \text{F}, \text{FS}, \text{FSC}, \text{FC}, \text{FCS}\}$. This full inventory reflects the broader experimental vocabulary retained by the production model.

5.2 Meaning Functions

Context-dependent size semantics. We model size adjectives with a context-sensitive threshold semantics following Schmidt, Goodman, Barner, and Tenenbaum (2009), who showed that the interpretation of gradable adjectives depends on the statistics of the comparison class. Franke et al. (2019) adopted the same approach in their simulation of adjective-ordering preferences. A threshold is computed from the current comparison class $\mathcal{C}$:

$$\theta_k(\mathcal{C}) = x_{\max}(\mathcal{C}) - k(x_{\max}(\mathcal{C}) - x_{\min}(\mathcal{C})), \quad (1)$$

where $x_{\min}(\mathcal{C})$ and $x_{\max}(\mathcal{C})$ are the lower and upper bounds of the mid-range of the contextual size distribution and $k \in [0, 1]$ controls how permissively “big” is applied. The resulting semantic value $\llbracket S \rrbracket(o, \mathcal{C}) \in [0, 1]$ is graded rather than binary: an object’s size is compared to the threshold via a normal slack function scaled by a perceptual-blur parameter w_f , consistent with scalar variability in the Approximate Number System (Feigenson, Dehaene, & Spelke, 2004). This context dependence is what makes the context-fixed vs. context-updating distinction possible: because θ_k depends on $\mathcal{C}$, changing which objects are in the comparison class changes what counts as big.

Colour and form semantics. Colour and form are in principle determinate: an object either matches the feature or it does not. We follow the relaxed-semantics approach of Degen et al. (2020) and Waldon and Degen (2021) by softening these denotations with $\nu \in (0, 1)$, which can be interpreted as the probability that a feature-matching object is correctly described by the adjective. This relaxation allows even determinate properties to carry graded information, which is needed for modelling redundant modification. In the reported model fits, colour is the primary determinate comparison case because inference is carried out on size–colour scenes, whereas form is retained as a general determinate feature term and as a possible redundant modifier in production. In the slider models, colour and form share a single ν parameter. In the production models, they receive separate fixed values, ν_{colour} and ν_{form} , whose provenance and sensitivity checks are summarized in Appendix D; the values are held constant across all final model variants.

5.3 Incremental Listener

The literal listener constructs utterance meanings incrementally by combining token-level semantic contributions while updating the effective comparison class. Let $u = (w_1, \dots, w_T)$ be an utterance. The listener maintains a sequence of belief states $\{P_t(o)\}_{t=0}^T$, where $P_t(o)$ denotes the posterior belief after incorporating the suffix $(w_{t+1}, \dots, w_T)$:

$$P_t(o) = P(o \mid w_{t+1}, \dots, w_T), \quad t = 0, \dots, T, \quad (2)$$

with base case

$$P_T(o) = P_{\text{prior}}(o). \quad (3)$$

For $t = T, T-1, \dots, 1$, beliefs are updated according to

$$P_{t-1}(o) \propto P_t(o) \llbracket w_t \rrbracket(o), \quad (4)$$

which normalizes to

$$P_{t-1}(o) = \frac{P_t(o) \llbracket w_t \rrbracket(o)}{\sum_{o'} P_t(o') \llbracket w_t \rrbracket(o')}. \quad (5)$$

After all tokens have been processed, the listener arrives at

$$P_0(o) = P(o \mid u). \quad (6)$$

The reverse indexing reflects prenominal composition: material closer to the noun establishes the comparison class for material farther from the noun. The composition rule in Equation (4) is shared by both semantic regimes of the model. What differs is how the meaning function $\llbracket w_t \rrbracket$ is computed for gradable adjectives such as size.

Context-updating semantics. Under the context-updating regime, the size threshold θ_k is recomputed from the listener’s current belief state at each composition step:

$$\llbracket w_t \rrbracket_{\text{UPD}}(o) = \llbracket w_t \rrbracket_{P_t}(o), \quad (7)$$

where P_t is the posterior after processing all words to the right of position t . The current belief state therefore plays a dual role: it serves as the prior for the next update step, and it

determines the semantics of the next token. Because P_t itself depends on the semantics of later words, this creates a feedback loop:

$$\llbracket w_t \rrbracket_{P_t} \text{ depends on } P_t(o) \propto P_{t+1}(o) \llbracket w_{t+1} \rrbracket_{P_{t+1}}(o). \quad (8)$$

Word order changes the sequence of comparison classes and thereby the interpretation of the utterance, yielding non-commutativity:

$$P(o \mid w_1, w_2) \neq P(o \mid w_2, w_1) \text{ in general.} \quad (9)$$

Context-fixed semantics. Under the context-fixed regime, the size threshold is computed once from the scene prior and held constant throughout composition:

$$\llbracket w_t \rrbracket_{\text{FIX}}(o) = \llbracket w_t \rrbracket_{P_{\text{prior}}}(o) \text{ for all } t. \quad (10)$$

The listener’s beliefs still update incrementally via Equation (4), but this update no longer feeds back into the meaning function. For deterministic adjectives such as colour and form, both regimes are identical because their semantics does not depend on the comparison class. The context-fixed regime therefore isolates the contribution of incremental belief updating from the additional effect of posterior-dependent meaning.

5.4 Utterance Prior and Speaker Models

The model incorporates a prior over utterance forms that is separate from the pragmatic component. This prior captures stable ordering tendencies — such as the well-documented subjectivity-driven size-first preference — that the speaker brings to the task before any scene-specific pragmatic reasoning.

In the slider-rating models, the prior is minimal: a single inferred bias parameter $b \geq 0$ that penalizes the reverse (colour-first) order relative to the canonical (size-first) order. Because b is estimated from data rather than fixed, the posterior indicates how much residual ordering preference remains after the RSA mechanism has been applied; if the pragmatic model were sufficient on its own, the posterior for b would concentrate near zero.

In the production models, the prior is derived from a language-model distribution over the 15 utterance forms, estimated from German GPT-2 surprisal values averaged across the experimental adjective–noun combinations. To keep this prior from explaining away utterance length or adjective content, we use only its residual order preference within each unordered content set. For example, the prior can distinguish SC from CS, but it cannot by itself prefer two-adjective over one-adjective descriptions, or colour descriptions over size descriptions. The contribution of this order-only prior is scaled by an inferred parameter β . Like b in the slider model, it encodes stable word-order tendencies from language experience, which the pragmatic speaker then modulates in light of the current scene. Against this shared listener and prior, the two speaker architectures differ in when pragmatic utility is evaluated.

Global speaker. The global speaker computes a distribution over complete utterances by applying a softmax to utterance-level utility:

$$S_{\text{gb}}(u \mid o) = \frac{\exp(U(o, u))}{\sum_{u'} \exp(U(o, u'))}. \quad (11)$$

For the shared production core, utility is

$$U(o^*, u) = \alpha \log L(o^* \mid u) - \beta \Omega(u), \quad (12)$$

where α is the RSA rationality parameter and $\Omega(u)$ is the order-only language-model score. The production model adds the salience, stopping, and lapse terms described in Section 6.2 to this core utility. In the slider-rating models, the utility simplifies to

$$U(o, u) = \alpha (\log L(o \mid u) - b_u), \quad (13)$$

where $b_u = 0$ for the size-first order and $b_u = b$ for the reverse order.

Incremental speaker. The incremental speaker chooses one token at a time. For an intended referent o^* ,

$$S_{\text{inc}}(u \mid o^*) = \prod_{t=1}^T S_t(w_t \mid u_{<t}, o^*), \quad u_{<t} = (w_1, \dots, w_{t-1}), \quad (14)$$

with local decision rule

$$S_t(a \mid u_{<t}, o^*) \propto [L(o^* \mid u_{<t} + a)]^\alpha. \quad (15)$$

Combining local pragmatic choice with the order-only prior yields the shared incremental production core:

$$S_{\text{inc}}(u \mid o^*) \propto \exp\{\beta \Omega(u)\} \cdot \prod_{t=1}^T \frac{[L(o^* \mid u_{<t} + w_t)]^\alpha}{Z_t(u_{<t}, o^*)}, \quad (16)$$

where

$$Z_t(u_{<t}, o^*) = \sum_a [L(o^* \mid u_{<t} + a)]^\alpha \quad (17)$$

ensures local normalization. In the slider-rating models, the LM prior is replaced by the bias term: $S_{\text{inc}}(u \mid o^*) \propto \exp(-b_u) \cdot \prod_t S_t(w_t \mid u_{<t}, o^*)$. The theoretical contrast is that the global speaker applies pragmatic reasoning once over complete utterances, whereas the incremental speaker applies it at each token position. When semantics is context-dependent, these two model variants are not equivalent. The equations in this subsection define the shared pragmatic core. The constrained-production fits then apply the same production extensions — order-only prior, visual-salience terms, salience-sensitive stopping cost, and lapse mixture — in every model of the production 2×2 .

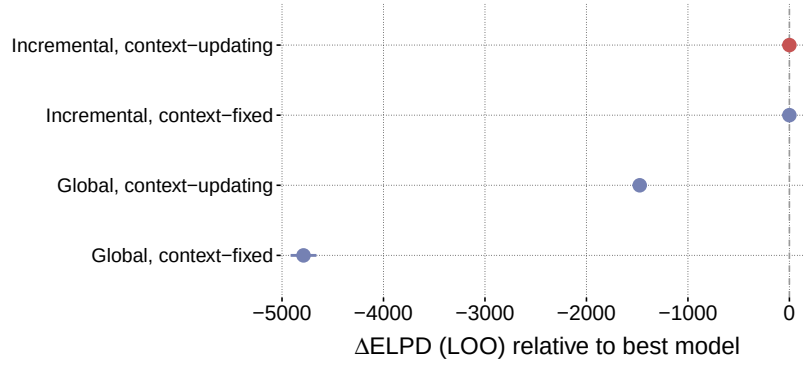
6 Bayesian Model Comparison

The 2×2 comparison crosses the two mechanisms introduced in Section 3.2: speaker architecture (global vs. incremental) and semantic regime (context-fixed vs. context-updating). The comparison is fitted independently to both datasets with hierarchical participant pooling. Within each dataset, the four models use matched parameter inventories. Fit differences are therefore interpreted as evidence about speaker architecture and semantic regime rather than as a consequence of unequal expressive flexibility, subject to the diagnostic caveats reported below and in the appendix. The comparison is diagnostic in three ways. A consistent advantage for incremental over global speakers would indicate that word-by-word choice contributes predictive accuracy beyond utterance-level choice. A consistent advantage for context-updating over context-fixed semantics would indicate that

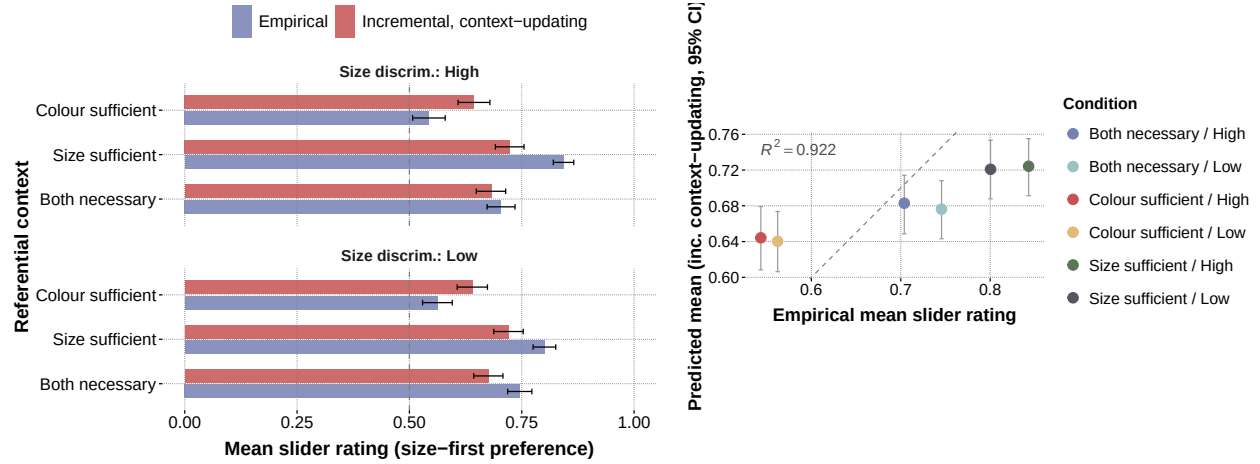
recomputing gradable meanings from the compositionally updated comparison class contributes predictive accuracy beyond incremental belief updating alone. A positive difference-in-differences would indicate that the value of context-updating semantics depends on whether production itself is incremental. We use Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO; Vehtari et al., 2017). For each comparison, we report differences in expected log pointwise predictive density (ΔELPD) relative to the best-fitting model together with the difference standard error (dse). A ratio $|\Delta\text{ELPD}|/\text{dse}$ substantially greater than 1 is treated as evidence for a clear difference in out-of-sample predictive accuracy. All models were implemented in NumPyro (Phan, Pradhan, & Jankowiak, 2019) and estimated with four MCMC chains using the NUTS sampler (Hoffman & Gelman, 2014). All models reported in the main text use hierarchical participant-level pooling, implemented as participant-level offsets on the rationality parameter. The incremental production models that drive the main comparison converged cleanly; diagnostics for the global robustness check are reported in Appendix C. The inferential focus is the comparison among incremental models, because these models both fit the data well and have acceptable MCMC diagnostics. Global models are retained as structurally informative baselines, but when they combine very poor predictive fit with weak diagnostics, their parameter estimates should not be interpreted substantively. Appendix C reports the available PSIS-LOO warnings, convergence summaries, and production Pareto- k counts for the main comparisons. Dataset-specific parameter inventories, likelihoods, and sampler settings are introduced with each subsection below.

6.1 Slider-Rating Study

The slider 2×2 is fitted on a minimal speaker with a single rationality parameter α , a bias intercept b , and a participant-level offset, with the semantic constants held fixed at $k = 0.5$, $w_f = 1.0$, and $\nu = 0.80$. Slider responses cluster at the boundaries (0 and 1) and are modelled with a Zero-One-Inflated Beta likelihood; each chain drew 500 warmup and 500 post-warmup samples (target acceptance probability = 0.90, maximum tree depth = 8).

**Figure 5**

Slider-rating model comparison: 2×2 speaker $\times$ semantics. Δ ELPD relative to the best model, with ± 1 difference standard error.

**Figure 6**

Posterior-predictive fit of the incremental, context-updating slider model. (a) Observed mean slider ratings and posterior-predictive model means by referential context $\times$ size discriminability; dashed line marks 0.5 (no preference), error bars are 95% confidence and posterior-predictive intervals. (b) Observed vs. predicted condition means; dashed line is identity.

Figure 5 shows the 2×2 comparison for the slider data. The largest contrast is speaker architecture: both incremental models fit the data well and are statistically

indistinguishable ($\Delta\text{ELPD} = 0.06$, $\text{dse} = 0.18$, $|\Delta\text{ELPD}|/\text{dse} = 0.32$), whereas both global models perform far worse ($\Delta\text{ELPD} > 1,400$). Within the incremental pair, context-updating semantics confers no detectable advantage over context-fixed semantics. For graded preference judgments between two competing orders, the predictive gain is carried by incremental pragmatic evaluation. This result does not show that posterior-dependent size meanings are absent; it shows that, within the incremental slider models, recomputing the size threshold from the running posterior does not improve out-of-sample prediction. Posterior-predictive checks show that the incremental context-updating model captures the condition-level ordering well, tracking the empirical ordering across all six referential context $\times$ size-discriminability combinations with $r = 0.96$ (squared Pearson ≈ 0.92 ; Figure 6). The model slightly under-predicts the strongest size-first conditions (size-sufficient) and slightly over-predicts the weakest (colour-sufficient), compressing the predicted range relative to the empirical range while preserving the rank-ordering. For the best-fitting incremental context-updating model, the posterior rationality parameter was $\alpha = 0.69$ ($SD = 0.06$, 94% HDI [0.57, 0.79]). The implied Beta concentration of the continuous ZOIB component was $\varphi = 1/\sigma^2 \approx 1.93$ (94% HDI [1.85, 2.04]), corresponding to $\text{Beta}(\mu \cdot 1.93, (1 - \mu) \cdot 1.93)$. This value implies a broad, near-uniform spread around the conditional mean, placing most of the conditional structure in μ and in the boundary inflations. The hierarchical spread of participant-level rationality offsets was $\tau = 0.12$ ($SD = 0.01$, 94% HDI [0.10, 0.14]). The residual bias intercept was $b = 0.58$ ($SD = 0.06$, 94% HDI [0.47, 0.69]), with posterior probability 0.91 above 0.50. This positive bias means that even after the RSA mechanism accounts for context-dependent informativeness, the model infers a residual size-first preference — consistent with the view that a stable ordering default persists beyond what discriminatory utility alone explains. The binary slider response space does not strongly discriminate between sharper and softer colour predicates: the slider posterior for ν is broad (0.63, 94% HDI [0.50, 0.82]), places substantial mass near the lower prior boundary at 0.5, only just excludes 0.80, and the four models remain statistically indistinguishable in

PSIS-LOO across this range (all $|\Delta\text{ELPD}|/\text{dse} < 1$; full numbers in Appendix F). The residual-bias estimate is conditional on the fixed $\nu = 0.80$ specification: under softer colour predicates the rationality α and the bias b shift substantially (towards 1.5 and below 0.5, respectively). Within the global pair, context-updating semantics substantially outperforms context-fixed semantics ($\Delta\text{ELPD} \approx 3,313$, $|\Delta\text{ELPD}|/\text{dse} > 24$). Context-updating composition thus provides a minimal source of order sensitivity that partially compensates for the lack of incrementality, but both global models remain far worse than either incremental variant.

6.2 Constrained-Production Study

Constrained production is a fifteen-way choice over one-, two-, and three-adjective strings, so the production speaker requires more than the binary bias used for slider ratings. At the same time, the production model must not encode each empirical regularity with a separate condition-specific parameter. We therefore use a parsimonious speaker with six population parameters and three interpretable components. These components instantiate the division of labor introduced in Section 3.2: stable order expectations, non-pragmatic visual prominence, and listener-oriented utility are represented separately and held constant across all four models of the architecture $\times$ semantics comparison. No component directly encodes referential context as a condition label; size discriminability enters through display-derived quantities such as target–competitor size contrast and the sharpness adjustment in the salience term.

Principled production components. First, stable ordering preferences enter through the order-only language-model score $\Omega(u)$ introduced above. Because $\Omega(u)$ is residualized within unordered content sets, it contributes a baseline ordering preference without carrying independent information about content choice or utterance length. Second, each adjective receives a soft visual-salience score. The base salience order is colour $>$ form $>$ size, reflecting the salience and consistency of colour in referring-expression production relative to size (Tarenskeen, Broersma, & Geurts, 2015), but the score is adjusted by the

current display. Let $\eta = (0, 1, 0.25)$ be the fixed base salience vector for size, colour, and form. For each trial, we compute a context adjustment

$$m = \begin{pmatrix} \sigma\left(\frac{s_0 - \max_{i>0} s_i}{\text{sd}(s) + 10^{-8}}\right) (0.5 + 0.5 \mathbb{I}_{\text{sharp}}) \\ 1 - \frac{1}{n-1} \sum_{i>0} \mathbb{I}[c_i = c_0] \\ 1 - \frac{1}{n-1} \sum_{i>0} \mathbb{I}[f_i = f_0] \end{pmatrix}, \quad (18)$$

where σ is the logistic sigmoid, the first term increases with target–competitor size contrast, and the latter two terms increase when fewer distractors share the target colour or form.

The trial-specific salience vector is then centered,

$$z = \eta + m - \text{mean}(\eta + m), \quad (19)$$

because only relative salience differences affect the softmax. Salience enters twice. In the incremental speaker it boosts locally salient choices in the token-level decision rule,

$$S_t(a \mid u_{<t}, o^*) \propto \exp\{\alpha \log L(o^* \mid u_{<t} + a) + \lambda_{\text{salience}} z_a\}, \quad (20)$$

For the global production speaker, the same salience scores enter as an utterance-level sum over the adjectives present in the string, so the global models receive the same non-pragmatic salience support without word-by-word choice. In both architectures, salience also imposes a stopping pressure when a salient adjective would be followed by additional material,

$$\text{cost}_{\text{stop}}(u) = \rho_{\text{stop}} \sum_{t < T} \max(z_{a_t}, 0). \quad (21)$$

This captures the tendency for highly salient sufficient colour to appear in short colour-only responses without hard-coding the colour-sufficient condition. Third, a single rationality parameter α controls pragmatic sensitivity to the listener’s posterior, and a small lapse rate ϵ accounts for residual unsystematic choices. The participant hierarchy allows individual differences in pragmatic strength through a shared scale τ .

The determinate feature semantics are held fixed at $\nu_{\text{colour}} = 0.59$ and $\nu_{\text{form}} = 0.50$, with $k = 0.50$ and $w_f = 0.69$ for size. These values are used uniformly across the production

comparison. The resulting model has six population parameters: α , $\log \beta$, $\lambda_{\text{salience}}$, ρ_{stop} , ϵ , and τ . The uncertainty-based length term considered during model development is not part of the final production model; the final fit is driven by the order-only prior, soft salience, pragmatic informativeness, and salience-sensitive stopping. Appendix D summarizes the model-development checks that led to this inventory.

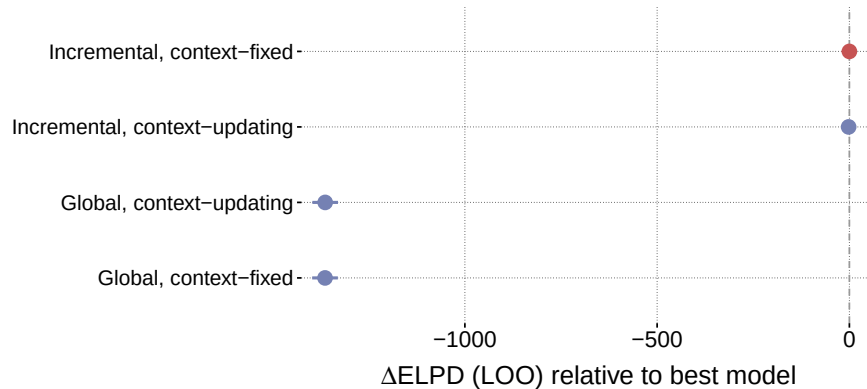
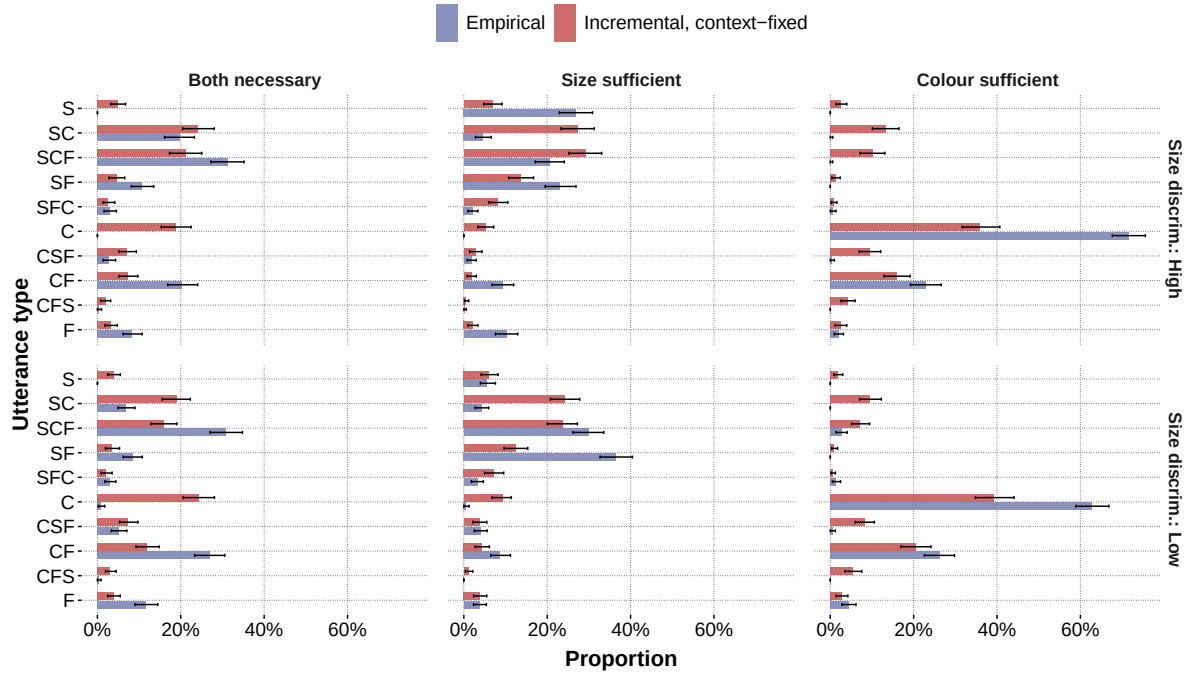


Figure 7

Constrained-production model comparison: PSIS-LOO ΔELPD relative to the best model for the parameter-matched 2×2 (speaker $\times$ semantics), with ± 1 difference standard error.

The four production models are parameter-matched on this inventory: identical six-parameter set, identical participant hierarchy, and identical constants. Predictive differences are therefore interpreted as reflecting architecture and semantic regime rather than expressive flexibility. Production responses are categorical over the fifteen utterance types; each chain drew 4,000 warmup and 2,000 post-warmup samples (target acceptance probability = 0.85, maximum tree depth = 7). The best-fitting production model is the incremental, context-fixed variant. It captures the qualitative production pattern while remaining deliberately conservative in fit: across the 90 condition $\times$ utterance combinations, observed and posterior-predictive proportions correlate at $r = 0.73$ ($R^2 = 0.53$; Figure 8 shows the by-condition fit and Figure 9 the condition-level correlation).

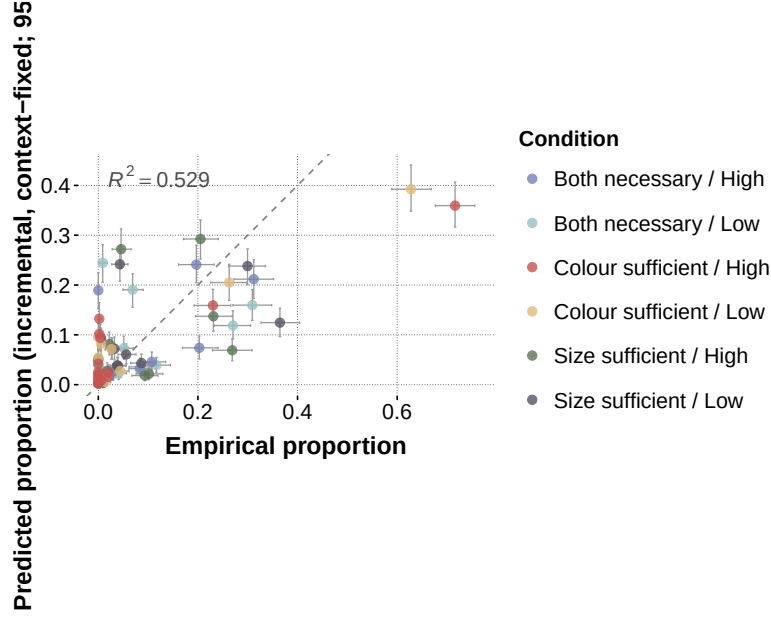
The largest predictive difference is between speaker architectures. Both incremental models outperform both global models by more than 1,300 ELPD in the matched

**Figure 8**

Posterior-predictive utterance-type proportions for the incremental, context-fixed production model. Observed proportions and posterior-predictive model proportions are shown by referential context $\times$ size discriminability; error bars are 95% bootstrap and posterior-predictive intervals.

comparison, and the global models show much weaker condition-level correlations ($r \approx 0.54$). The production comparison therefore supports word-by-word speaker choice over complete-utterance choice as a source of predictive accuracy. This conclusion is about predictive fit; the global models are retained as poor-fitting baselines rather than as sources of substantively interpretable parameter estimates.

Within the incremental pair, the context-fixed model is favoured by PSIS-LOO. The context-fixed model is the best-fitting production model ($\text{ELPD}_{\text{LOO}} = -6,367.34$, $p_{\text{LOO}} = 49.01$), followed closely by the context-updating model ($\text{ELPD}_{\text{LOO}} = -6,369.32$, $\Delta\text{ELPD} = 1.98$, $\text{dse} = 0.37$). The difference is small relative to the architecture effect but precise at the scale of the incremental-pair comparison, and it is in the direction of the

**Figure 9**

Observed vs. predicted utterance-type proportions across all condition $\times$ utterance combinations for the incremental, context-fixed production model; dashed line is identity.

simpler semantic regime. The fitted production comparison therefore provides no evidence for an additional predictive benefit from context-updating semantics. The incremental advantage is robust, but posterior-dependent size meanings do not improve out-of-sample prediction once the final production model includes the order-only prior, display-sensitive salience, salience-sensitive stopping, and lapse mixture. The same conclusion holds in a robustness check in which the global lapse rate is fixed at the incremental posterior value, eliminating the unstable lapse-rationality trade-off in the global models; the difference-in-differences remains negative (-2.30 ELPD), and no model has problematic Pareto- k values.

The posterior parameters of the incremental, context-fixed model are reported in full in Table B1. The posterior for the order-prior weight is clearly positive ($\log \beta \approx 0.59$), but it does not account for the fit by itself. Pragmatic rationality is positive ($\alpha \approx 1.44$), and both salience parameters are clearly identified ($\lambda_{\text{salience}} \approx 1.57$, $\rho_{\text{stop}} \approx 0.49$). The lapse rate is near zero ($\epsilon \approx 0.003$), so the model is not relying on uniform noise to absorb misfit. The

participant-level spread is substantial but well behaved ($\tau \approx 0.84$), indicating stable individual variation in pragmatic strength.

6.3 Discussion

Comparing the two datasets yields a consistent architecture result. Both datasets favour incremental choice over global utterance evaluation in predictive fit, but they support that conclusion in different response spaces. In the slider data, the two incremental semantic regimes are statistically indistinguishable ($|\Delta\text{ELPD}|/\text{dse} = 0.32$). In the production data, the two incremental regimes are close relative to the architecture effect, but the context-fixed model is favoured by a small and precise difference. Thus the empirical comparison does not support the expected additional benefit of context-updating semantics in production. Context-updating composition still has a defined role in the mechanism space: the simulation asks what it can generate on its own, and it supplies order sensitivity for a global speaker without an order prior. In the fitted production comparison, that role does not translate into best predictive performance once the model includes an order-only prior, visual salience, and incremental choice. The response-space asymmetry is instead reflected in what each task identifies. The binary slider response space is informative about incremental pragmatic evaluation but weakly identifies the sharpness of the colour predicate; the production response space exposes a richer pattern of one-, two-, and three-adjective choices, making the incremental production process more diagnostic.

7 Simulation Study

7.1 Goals and Design

The fitted model comparison identifies which mechanisms add predictive accuracy, but it does not by itself determine what each mechanism can generate. This leaves an interpretive question for context-updating semantics. In the fitted comparisons, context-updating semantics adds little or no predictive accuracy within the best incremental models. The same mechanism may still be sufficient to produce a baseline size-first asymmetry in a generative model. The simulation addresses this question by applying the

same 2×2 crossing outside the fitted empirical models.

We ask three questions. First, can the crossed models *generate* a size-first ordering preference across a broad range of contexts without being fit to observed responses? Second, when do they generate colour-first reversals? Third, how do speaker architecture and semantic regime combine: does the baseline effect require context-updating semantics, or can it arise from incremental choice alone?

To answer these questions, we ran a parameter-grid simulation over the values listed in Table 3, crossing both speaker models (incremental, global) with both semantic regimes: context-updating (recursive; Equation (7)) and context-fixed (static; Equation (10)). Under context-updating semantics, the size threshold adapts to the literal listener’s running posterior at each composition step; under context-fixed semantics, the threshold is computed once from the uniform prior and held fixed. For each parameter combination, 1,000 random contexts were sampled at the specified context size, yielding 12,000,000 simulated trials in total. On each trial, the simulation computed the pragmatic listener’s posterior probability of the target referent after the size-first utterance “big blue” and after the colour-first utterance “blue big”. Speaker architecture enters through this pragmatic listener: L_1 is obtained by Bayesian inversion of either a global or an incremental speaker, so the simulation measures the communicative success of fixed utterances under different speaker architectures rather than fitted utterance frequencies. The simulation estimand is the size-first advantage:

$$\Delta_{\text{SF}} = L_1(o^* \mid \text{“big blue”}) - L_1(o^* \mid \text{“blue big”}).$$

Positive values indicate that the size-first order has higher communicative success; negative values indicate a colour-first advantage. The plotted values are averages of this quantity over sampled contexts and parameter settings within each displayed bin.

The parameter σ_{spread} controls the spread of the size distribution in the sampled context: low values produce low-discriminability contexts in which size is weakly informative, whereas high values produce high-discriminability contexts in which size strongly separates the target from its competitors. The remaining parameters (k , w_f , ν) instantiate the

Table 3*Simulation parameter grid*

Parameter	Grid values	# levels
Speaker model	Incremental, global	2
Semantics	Context-updating, context-fixed	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
ν	0.90, 0.92, 0.94, 0.96, 0.98	5
k	0.10, 0.30, 0.50, 0.70, 0.90	5
w_f	0.1, 0.2, 0.3, 0.5, 0.8	5
Random context samples per combination	1,000	
Total trials	12,000,000	

semantic space explored in the simulation. Because the goal is to test the qualitative consequences of semantic regime, the ν grid focuses on high-reliability determinate predicates, matching the theoretical regime in which colour and form are nearly categorical. The production model, by contrast, uses the empirically calibrated softer colour value $\nu_{\text{colour}} = 0.59$ for predictive fitting. The simulation is therefore a theoretical parameter-grid study rather than a posterior-predictive simulation of the fitted production model. The individual effects of the semantic grid parameters are reported in Appendix E.

7.2 Results

Figure 10 shows the size-first advantage as a function of context size and contextual spread, faceted by speaker model (columns) and semantic regime (rows).

The simulation yields three results that qualify the model-comparison interpretation. First, context-updating semantics can generate a size-first advantage without incremental

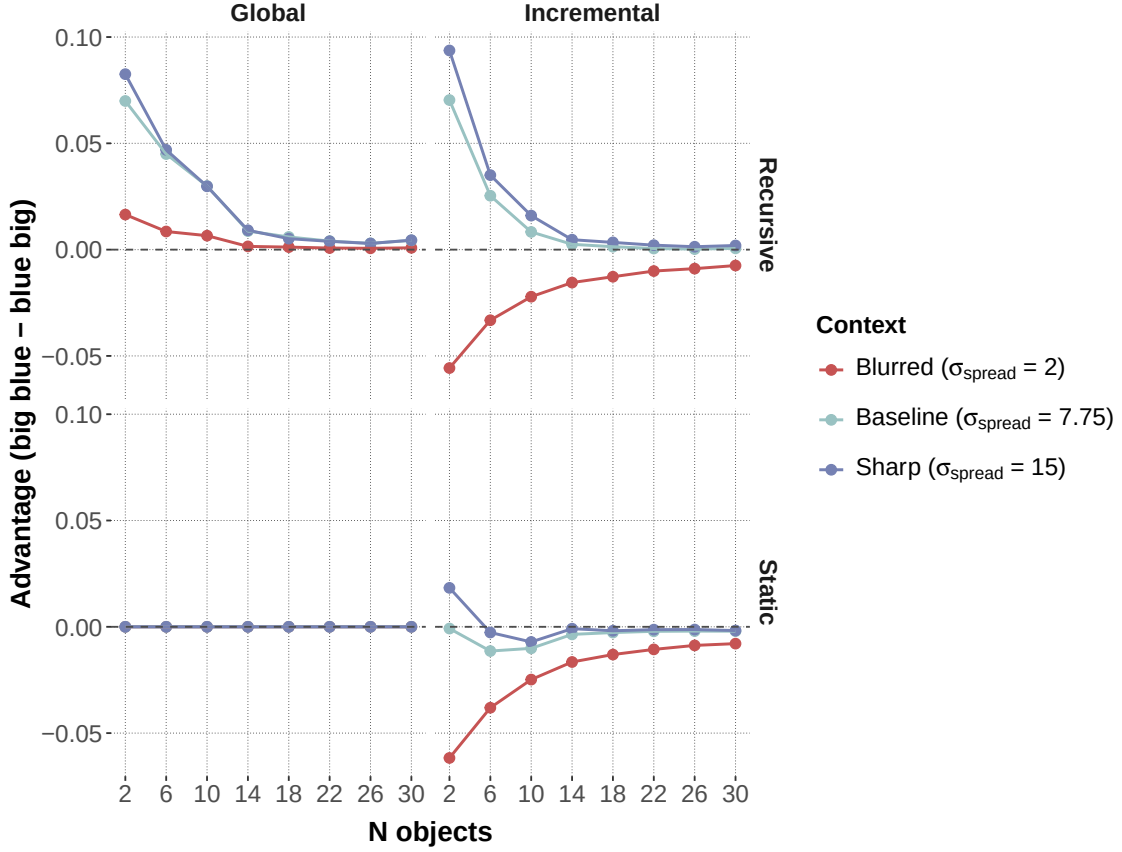


Figure 10

Simulation size-first advantage $\Delta_{SF} = L_1(o^* \mid \text{"big blue"}) - L_1(o^* \mid \text{"blue big"})$ as a function of context size (n_{obj}) and contextual spread (σ_{spread}), faceted by speaker model (columns) and semantic regime (rows). Positive values indicate higher communicative success for the size-first order.

speaker choice. Under context-updating semantics, both speaker models generate a size-first advantage across most of the parameter space. A global speaker with context-sensitive composition therefore derives the canonical ordering asymmetry without an incremental production policy. This result is consistent with the subjectivity literature: once semantic values are evaluated over progressively restricted comparison classes, less noisy inner modification can produce the familiar ordering direction (Franke et al., 2019; Scontras et al., 2020, 2019).

Second, context-fixed semantics removes this global speaker advantage. Under context-fixed semantics, the global speaker produced $\Delta = 0.000$ across the parameter grid. When the size threshold does not update during composition, the literal listener assigns identical posteriors to both orderings, making them informationally equivalent. The global speaker’s baseline advantage therefore comes from order-sensitive threshold adaptation rather than from utterance-level choice.

Third, incremental choice changes the shape of the ordering pattern. The incremental speaker strengthens the size-first advantage under context-updating semantics in much of the parameter space, and it also produces colour-first preferences in low-discriminability, low-cardinality contexts. Under context-fixed semantics, the incremental speaker defaults slightly to colour-first ($\Delta = -0.009$), driven by the chain-rule asymmetry that favours “blue” as the opening token when size is insufficiently diagnostic. The simulation therefore separates two sources of order sensitivity: context-updating composition can provide a baseline asymmetry, and incremental choice determines how adjective order responds to the relative informativeness of each adjective at the point where it is selected.

8 General Discussion

finish this

9 Data Availability Statement

Data, analysis scripts, model code, and figure-generation scripts will be made available in an anonymized repository upon submission. The repository is intended to contain the cleaned experimental data, preprocessing scripts, NumPyro model code, posterior summaries, and manuscript figure scripts needed to reproduce the reported analyses.

10 Author Note

Acknowledgments, funding information, author contributions, and competing-interest statements will be finalized with the author team before journal submission.

11 References

- Cinque, G. (1994). On the evidence for partial N-movement in the Romance DP. In G. Cinque, J. Koster, J.-Y. Pollock, L. Rizzi, & R. Zanuttini (Eds.), *Paths towards universal grammar: Studies in honor of Richard S. Kayne* (pp. 85–110). Washington, DC: Georgetown University Press.
- CohnGordon, R., Goodman, N. D., & Potts, C. (2019). An incremental iterated response model of pragmatics. In *Proceedings of the society for computation in linguistics* (Vol. 2, pp. 81–90). doi: 10.7275/7yvv-jt08
- Degen, J. (2023). The rational speech act framework. *Annual Review of Linguistics*, 9, 519–540.
- Degen, J., Hawkins, R. D., Graf, C., Kreiss, E., & Goodman, N. D. (2020). When redundancy is useful: A bayesian approach to “overinformative” referring expressions. *Psychological Review*, 127(4), 591.
- Dixon, R. M. W. (1982). *Where have all the adjectives gone? and other essays in semantics and syntax*. Berlin: Mouton.
- Dyer, W., Torres, C., Scontras, G., & Futrell, R. (2023). Evaluating a century of progress on the cognitive science of adjective ordering. *Transactions of the Association for Computational Linguistics*, 11, 1185–1200.
- Feigenson, L., Dehaene, S., & Spelke, E. (2004). Core systems of number. *Trends in Cognitive Sciences*, 8(7), 307–314.
- Franke, M., Scontras, G., & Simoncic, M. (2019). Subjectivity-based adjective ordering maximizes communicative success. In *Cogsci* (pp. 344–350).
- Fukumura, K. (2018). Ordering adjectives in referential communication. *Journal of Memory and Language*, 101, 37–50.
- Fukumura, K., Van Gompel, R. P., & Pickering, M. J. (2010). The use of visual context during the production of referring expressions. *Quarterly journal of experimental psychology*, 63(9), 1700–1715.

- Futrell, R., Dyer, W., & Scontras, G. (2020, July). What determines the order of adjectives in English? comparing efficiency-based theories using dependency treebanks. In D. Jurafsky, J. Chai, N. Schluter, & J. Tetreault (Eds.), *Proceedings of the 58th annual meeting of the association for computational linguistics* (pp. 2003–2012). Online: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2020.acl-main.181/> doi: 10.18653/v1/2020.acl-main.181
- Goodman, N. D., & Frank, M. C. (2016). Pragmatic language interpretation as probabilistic inference. *Trends in cognitive sciences*, 20(11), 818–829.
- Hahn, M., Degen, J., Goodman, N., Jurafsky, D., & Futrell, R. (2018). An information-theoretic explanation of adjective ordering preferences. In *Proceedings of the 40th annual meeting of the cognitive science society (cogsci)* (pp. 1766–1771). Retrieved from <https://cogsci.mindmodeling.org/2018/papers/0339/index.html>
- Hoffman, M. D., & Gelman, A. (2014). The no-u-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(47), 1593–1623.
- Martin, J. E. (1969). Semantic determinants of preferred adjective order. *Journal of Verbal Learning and Verbal Behavior*, 8(6), 697–704.
- Phan, D., Pradhan, N., & Jankowiak, M. (2019). Composable effects for flexible and accelerated probabilistic programming in numpyro. *arXiv preprint arXiv:1912.11554*.
- Quirk, R., Greenbaum, S., Leech, G., & Svartvik, J. (1972). *A grammar of contemporary english*. London: Longman.
- Rubio-Fernández, P. (2016). How redundant are redundant color adjectives? an efficiency-based analysis of color overspecification. *Frontiers in psychology*, 7, 153.
- Rubio-Fernandez, P. (2019). Overinformative speakers are cooperative: Revisiting the gricean maxim of quantity. *Cognitive science*, 43(11), e12797.

- Rubio-Fernandez, P., & Jara-Ettinger, J. (2020). Incrementality and efficiency shape pragmatics across languages. *Proceedings of the National Academy of Sciences*, 117(24), 13399–13404.
- Rubio-Fernández, P., Mollica, F., & Jara-Ettinger, J. (2021). Speakers and listeners exploit word order for communicative efficiency: A cross-linguistic investigation. *Journal of Experimental Psychology: General*, 150(3), 583–594.
- Schlotterbeck, F., & Wang, H. (2023). An incremental rsa model for adjective ordering preferences in referential visual context. *Proceedings of the Society for Computation in Linguistics*, 6(1), 121–132.
- Schmidt, L. A., Goodman, N. D., Barner, D., & Tenenbaum, J. B. (2009). How tall is tall? Compositionality, statistics, and gradable adjectives. In N. A. Taatgen & H. van Rijn (Eds.), *Proceedings of the 31st annual conference of the cognitive science society*.
- Scontras, G. (2023). Adjective ordering across languages. *Annual Review of Linguistics*, 9, 357–376.
- Scontras, G., Bar-Sever, G., Kachakeche, Z., Rosales Jr, C. M., & Samonte, S. (2020). Incremental semantic restriction and subjectivity-based adjective ordering. In *Proceedings of sinn und bedeutung* (Vol. 24, pp. 253–270).
- Scontras, G., Degen, J., & Goodman, N. D. (2017). Subjectivity predicts adjective ordering preferences. *Open Mind*, 1(1), 53–66.
- Scontras, G., Degen, J., & Goodman, N. D. (2019). On the grammatical source of adjective ordering preferences. *Semantics and Pragmatics*, 12, 7–1.
- Scott, G.-J. (2002). Stacked adjectival modification and the structure of nominal phrases. In G. Cinque (Ed.), *Functional structure in DP and IP: The cartography of syntactic structures, vol. 1* (pp. 91–120). Oxford: Oxford University Press.
- Sedivy, J. C., Tanenhaus, M. K., Chambers, C. G., & Carlson, G. N. (1999). Achieving incremental semantic interpretation through contextual representation. *Cognition*, 71(2), 109–147.

- Sproat, R., & Shih, C. (1991). The cross-linguistic distribution of adjective ordering restrictions. In C. Georgopoulos & R. Ishihara (Eds.), *Interdisciplinary approaches to language: Essays in honor of S.-Y. Kuroda* (pp. 565–593). Kluwer.
- Sweet, H. (1898). *A new english grammar, logical and historical, vol. 2: Syntax*. London: Clarendon.
- Tanenhaus, M. K., Spivey-Knowlton, M. J., Eberhard, K. M., & Sedivy, J. C. (1995). Integration of visual and linguistic information in spoken language comprehension. *Science*, 268(5217), 1632–1634.
- Tarenskeen, S., Broersma, M., & Geurts, B. (2015). Overspecification of color, pattern, and size: salience, absoluteness, and consistency. *Frontiers in Psychology*, 6, 1703.
- Teodorescu, A. (2006). Adjective ordering restrictions revisited. In D. Baumer, D. Montero, & M. Scanlon (Eds.), *Proceedings of the 25th West Coast Conference on Formal Linguistics* (pp. 399–407). Somerville, MA: Cascadilla Proceedings Project.
- Trainin, N., & Shetreet, E. (2021). It’s a dotted blue big star: on adjective ordering in a post-nominal language. *Language, Cognition and Neuroscience*, 36(3), 320–341.
- Trotzke, A., & Wittenberg, E. (2019). Long-standing issues in adjective order and corpus evidence for a multifactorial approach. *Linguistics*, 57(2), 273–282. doi: 10.1515/ling-2019-0001
- Van Gompel, R. P., Van Deemter, K., Gatt, A., Snoeren, R., & Krahmer, E. J. (2019). Conceptualization in reference production: Probabilistic modeling and experimental testing. *Psychological review*, 126(3), 345.
- Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5), 1413–1432.
- Waldon, B., & Degen, J. (2021). Modeling cross-linguistic production of referring expressions. In *Proceedings of the society for computation in linguistics 2021* (pp. 206–215). Association for Computational Linguistics.
- Whorf, B. L. (1945). Grammatical categories. *Language*, 21(1), 1–11.

Appendix A

Methods and Coding Transparency

Table A1 summarizes the participant exclusions and the analysis tables used in the reported manuscript analyses. The per-criterion exclusion counts are not mutually exclusive, so the unique number of excluded participants is smaller than the sum of the criterion counts. For Experiment 2, the broad preprocessed production table contains all three critical adjective-pair families. The production model comparison reported in Section 6.2 uses the size-colour subset with annotated responses, yielding 3,196 observations over the 15 utterance categories.

Table A1

Participant exclusions and analysis-table sizes. Criterion counts overlap within experiment.

Experiment	Starting and final N	Participant exclusions	Analysis table used in manuscript
Slider ratings	150 $\rightarrow$ 134	8 with more than three missing trial-list entries; 8 with extreme completion time; 2 with more than three control-item failures; 6 with more than three “none fits” responses on experimental items; 16 unique participants excluded.	Broad cleaned critical table: 10,606 observations. Reported size-colour slider subset: 3,485 observations.
Constrained production	149 $\rightarrow$ 113	8 with extreme completion time; 25 with more than 80% identical responses; 7 with more than 10% missing or unannotatable responses; 36 unique participants excluded.	Broad preprocessed critical table: 9,639 rows. Reported size-colour production model and figures: 3,196 annotated observations.

The slider-rating materials used the fixed adjective vocabulary in Table A2. The raw stimulus table encodes colours and forms with compact feature labels; the interface displayed the corresponding German adjective forms in phrases such as *den großen blauen Aufkleber*.

Table A2

Adjective vocabulary in the slider-rating item descriptions.

Property class	Raw stimulus values	German adjective forms displayed in the item text
Size	Numeric size levels 1–6, 9, 10	<i>großen, kleinen</i>
Colour	blue, orange, grey, brown, green, black	<i>blauen, orangen, grauen, braunen, grünen, schwarzen</i>
Form	circle, diamond, heart, quadrat, star, triangle	<i>runden, diamantförmigen, herzförmigen, quadratischen, sternförmigen, dreieckigen</i>

The preprocessed modelling table recodes the determinate contrasts to binary feature values, but this recoding is separate from the lexical forms shown to participants.

Production responses were coded into the 15 ordered adjective categories listed in Table A3. The data files use D for the size or dimension adjective, while the model exposition uses S; the two notations refer to the same adjective class. Misspellings and transparent lexical alternatives were normalized when their semantic class was unambiguous. Spatial descriptions, comparative or ordinal size expressions, and responses that could not be assigned to a target adjective class were excluded from the production model comparison.

Table A3

Production coding dictionary. In the data files, D corresponds to the size adjective S; C and F correspond to colour and form.

Length	Codes
One adjective	D, C, F
Two adjectives	DC, DF, CD, CF, FD, FC
Three adjectives	DCF, DFC, CDF, CFD, FDC, FCD

The descriptive analyses and the Bayesian model comparisons address different inferential questions. The descriptive regressions test the qualitative empirical patterns in the observed responses, whereas the Bayesian models compare predictive mechanisms under

response-appropriate likelihoods. For the reported size–colour slider analysis, the descriptive model was a linear mixed-effects model of centred slider rating with fixed effects of referential context, size discriminability, and their interaction, plus random intercepts for participant and item:

$$y_{\text{slider}} \sim \text{context} \times \text{discriminability} + (1 \mid \text{participant}) + (1 \mid \text{item}).$$

For production size-initiality, the descriptive model was a logistic mixed-effects model with a binary outcome indicating whether the coded response began with the size adjective:

$$y_{\text{size-initial}} \sim \text{context} \times \text{discriminability} + (1 \mid \text{participant}) + (1 \mid \text{item}).$$

For production over-informativity, the descriptive model used the same fixed and random structure with a binary outcome indicating a three-adjective response in both-necessary contexts or a multi-adjective response in one-property-sufficient contexts:

$$y_{\text{over-informative}} \sim \text{context} \times \text{discriminability} + (1 \mid \text{participant}) + (1 \mid \text{item}).$$

Appendix B

Production Model Details

Table B1 lists the population parameters of the advocated production model (Section 6.2), their priors, and their posterior summaries. The participant hierarchy adds 113 latent rationality offsets drawn from a zero-mean normal distribution with standard deviation τ . The fixed constants are $\nu_{\text{colour}} = 0.59$, $\nu_{\text{form}} = 0.50$, $k = 0.50$, and $w_f = 0.69$.

Table B1

Population parameters of the advocated production model: prior and posterior mean with 94% HDI.

Parameter	Role	Prior	Posterior
α	pragmatic rationality	HalfNormal(3)	1.44 [1.17, 1.68]
$\log \beta$	order-only LM-prior weight	Normal(0, 0.35)	0.59 [0.54, 0.65]
$\lambda_{\text{salience}}$	local boost for visually salient features	HalfNormal(1)	1.57 [1.50, 1.65]
ρ_{stop}	cost for continuing after a salient feature	HalfNormal(0.75)	0.49 [0.41, 0.59]
ϵ	lapse / uniform-noise rate	Beta(1, 50)	0.003 [0.000, 0.009]
τ	SD of participant rationality offsets	HalfNormal(0.15)	0.84 [0.69, 0.99]

Appendix C

Model Diagnostics and Production Robustness

Table C1 summarizes the available diagnostics for the main 2×2 comparisons. For the slider models, manuscript-facing summaries contain PSIS-LOO warnings and maximum split- $\hat{R}$, but they do not archive Pareto- k counts or ESS/divergence summaries. For the production models, the incremental models have clean MCMC diagnostics and no problematic Pareto- k observations; the global models fit much worse and have weak convergence, so their parameters are not interpreted substantively.

Table C1

Model-comparison diagnostics. Δ ELPD and dse are relative to the best model within dataset. Pareto- k counts are the number of observations with $k \geq .7$ when archived.

Model	Predictive summary	Diagnostics	Interpretation
Slider: incremental, updating	ELPD -2774.1 ; $p_{\text{LOO}} = 64.8$; $\Delta = 0.0$; dse $-$	PSIS warning yes; max $\hat{R} = 1.006$; Pareto- k n/a ; divergences n/a	Best slider model; incremental comparison trusted with PSIS warning noted.
Slider: incremental, fixed	ELPD -2774.1 ; $p_{\text{LOO}} = 65.1$; $\Delta = 0.06$; dse 0.18	PSIS warning yes; max $\hat{R} = 1.007$; Pareto- k n/a ; divergences n/a	Statistically indistinguishable from updating.
Slider: global, updating	ELPD -4249.2 ; $p_{\text{LOO}} = 1425.0$; $\Delta = 1475.1$; dse 65.9	PSIS warning yes; max $\hat{R} = 1.845$; Pareto- k n/a ; divergences n/a	Poor predictive baseline; parameters not interpreted.
Slider: global, fixed	ELPD -7562.2 ; $p_{\text{LOO}} = 4372.0$; $\Delta = 4788.1$; dse 127.2	PSIS warning yes; max $\hat{R} = 3.098$; Pareto- k n/a ; divergences n/a	Poor predictive baseline; parameters not interpreted.
Production: incremental, fixed	ELPD -6367.3 ; $p_{\text{LOO}} = 49.0$; $\Delta = 0.0$; dse $-$	PSIS warning no; $k \geq .7$: 0; max $\hat{R} = 1.00$; divergences 0	Best production model; main inferential reference.
Production: incremental, updating	ELPD -6369.3 ; $p_{\text{LOO}} = 48.6$; $\Delta = 1.98$; dse 0.37	PSIS warning no; $k \geq .7$: 0; max $\hat{R} = 1.00$; divergences 0	Clean diagnostics; close but lower than fixed semantics.
Production: global, updating	ELPD -7730.7 ; $p_{\text{LOO}} = 615.5$; $\Delta = 1363.4$; dse 33.3	PSIS warning yes; $k \geq .7$: 2; max $\hat{R} = 1.53$; divergences 0	Poor predictive baseline with weak convergence.
Production: global, fixed	ELPD -7731.1 ; $p_{\text{LOO}} = 615.8$; $\Delta = 1363.7$; dse 33.4	PSIS warning yes; $k \geq .7$: 1; max $\hat{R} = 1.54$; divergences 0	Poor predictive baseline with weak convergence.

The main production comparison uses PSIS-LOO on the matched six-parameter 2×2 . The two incremental models have clean diagnostics (maximum split- $\hat{R} = 1.00$, no divergences, and no Pareto- $k \geq .7$ observations). The global models are much worse fitting. Because global utterance-level choice can trade off strongly with the lapse term, we also checked a version in which the global lapse rate was fixed at $\epsilon = .003$, the posterior value of the advocated incremental model. This check leaves the substantive ordering of the models

unchanged: incremental context-fixed remains best, incremental context-updating remains second, and the difference-in-differences for the speaker $\times$ semantics contrast remains negative rather than positive (-2.30 ELPD). The fixed-lapse global check removes problematic Pareto- k values; global-chain convergence also improves substantially, though the global models remain less stable than the advocated incremental models (maximum split- $\hat{R} = 1.07$ for global context-updating and 1.05 for global context-fixed).

Appendix D

Production Model Development and Fixed-Constant Provenance

Tables D1 and D2 summarize the provenance of the fixed constants and production-model components used in the final matched comparison. These choices were not varied across the four models of the final 2×2 . They therefore define the common production speaker against which the architecture and semantic-regime contrasts are tested.

Table D1

Fixed constants used in the final matched comparison.

Constant	Value	Provenance or check	Claim status
Slider ν	0.80	Minimal slider-speaker specification for the size-colour slider data; free- ν audit reported in Appendix F.	Architecture conclusion unchanged; bias estimate is conditional on this fixed value.
Production ν_{colour}	0.59	Free-colour-semantic-value diagnostic on the size-colour production data; value re-fixed for cleaner matched fits.	Defines common production semantics; not itself a 2×2 contrast.
Production ν_{form}	0.50	Fixed soft determinate-feature constant used to retain form as a possible redundant modifier.	Form is retained for response-space coverage, not as the main mechanistic contrast.
Production k	0.50	Mid-range threshold default; simulation sweeps k in Appendix E.	Held fixed so models differ only in architecture and semantic regime.
Production w_f	0.69	Previous free- w_f diagnostic median from production model development; simulation sweeps w_f in Appendix E.	Held fixed in the production comparison.

The archived constant name for the production blur weight is

WF_FIXED_ITER11_MEDIAN.

Table D2*Production-component provenance for the final matched comparison.*

Component		Form		Provenance or check	Claim status
Order-only prior	LM	Residual	score	German GPT-2 surprisal residualized within unordered content sets; weight β inferred in every final model.	Retained; cannot explain length or content choice by construction.
Soft salience	visual	Centered	display score z_a	Fixed base vector plus display adjustment; no referential-context dummy; $\lambda_{\text{salience}}$ inferred.	Retained as non-pragmatic feature prominence.
Salience-sensitive stopping		Continuation	cost after salient words	Production model-development check on size-colour data; improves PSIS-LOO over the no-length variant by 52.57 ELPD (dse = 10.07).	Retained as short-description pressure.
Uncertainty-gated length term		Removed		Exploratory production check on size-colour data showed no clear benefit over the model without the term ($\Delta\text{ELPD} = 0.49$, dse = 0.88), and the posterior included zero.	Not part of the final production model.
Regularized hierarchy	hier-	Participant off-sets with conservative prior		Model-development check; slightly lower ELPD than the unregularized salience-stop fit ($\Delta\text{ELPD} \approx 2.16$, dse = 1.15) while reducing p_{LOO} .	Used for final matched comparison.

Table D3 summarizes the main production-model development checks. These checks were used to keep the final production model tied to independently motivated components while avoiding condition-specific residual parameters. The final 2×2 comparison uses the regularized salience-stop model because it retains the qualitative fit of the best exploratory variant while reducing effective complexity.

Table D3*Production-model development checks motivating the final six-parameter inventory.*

Component choice	or	Motivation	Check	Decision
Order-only LM prior		Stable order expectations without allowing the LM prior to explain content choice or utterance length.	LM scores residualized within unordered content sets, so the prior distinguishes SC from CS but not short from long descriptions.	Retained as $\Omega(u)$ with inferred weight β .
Soft visual salience		Feature prominence is independent of pragmatic information gain and should vary with the display.	Salience is context-dependent and centered within trial; no referential-context dummy is used.	Retained as local first-word boost $\lambda_{\text{salience}} z_a$.
Salience-sensitive stopping		Highly salient sufficient features should support short descriptions.	Adding the stopping term improves PSIS-LOO over the salience-only/no-length variant ($\Delta\text{ELPD} = 52.57$, $\text{dse} = 10.07$).	Retained as the continuation cost in Equation (21).
Uncertainty-gated length term		Blurry size could in principle motivate longer descriptions.	Initial comparison showed no clear benefit over the model without the term ($\Delta\text{ELPD} = 0.49$, $\text{dse} = 0.88$), and the posterior included zero.	Removed from the final model.
Regularized participant hierarchy		Participant variation is required, but the final architecture comparison should be conservative about effective complexity.	Regularization slightly lowers ELPD relative to the unregularized salience-stop fit ($\Delta\text{ELPD} \approx 2.16$, $\text{dse} = 1.15$) while reducing p_{LOO} from about 56.4 to 48.9.	Used for the final 2×2 comparison.

Appendix E

Simulation Parameter Sweeps

Figures E1 to E3 show how the three fixed semantic parameters affect communicative success across the two speaker models.

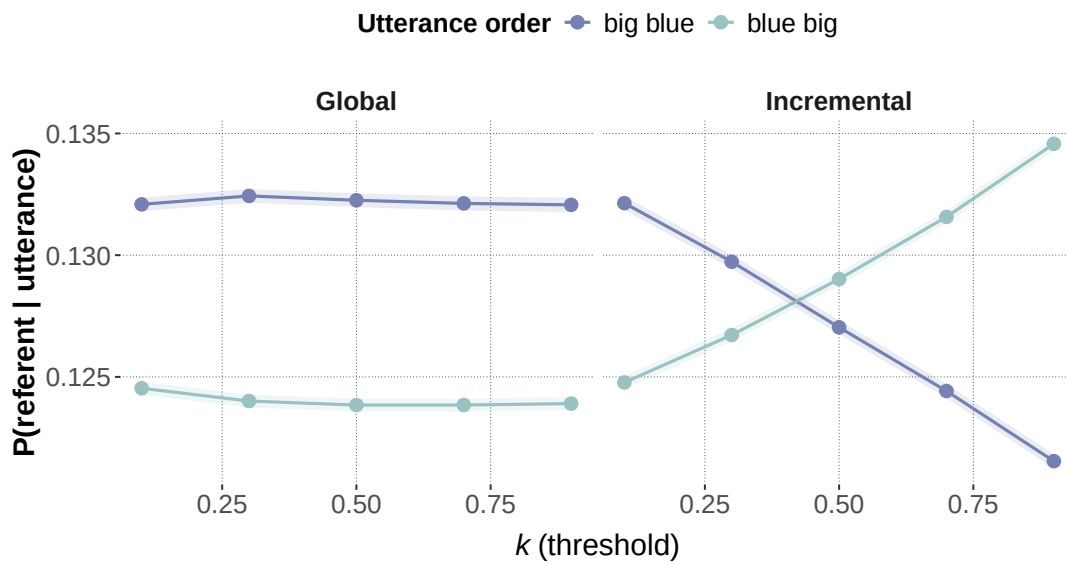


Figure E1

Effect of the threshold parameter k on communicative success in the simulation.

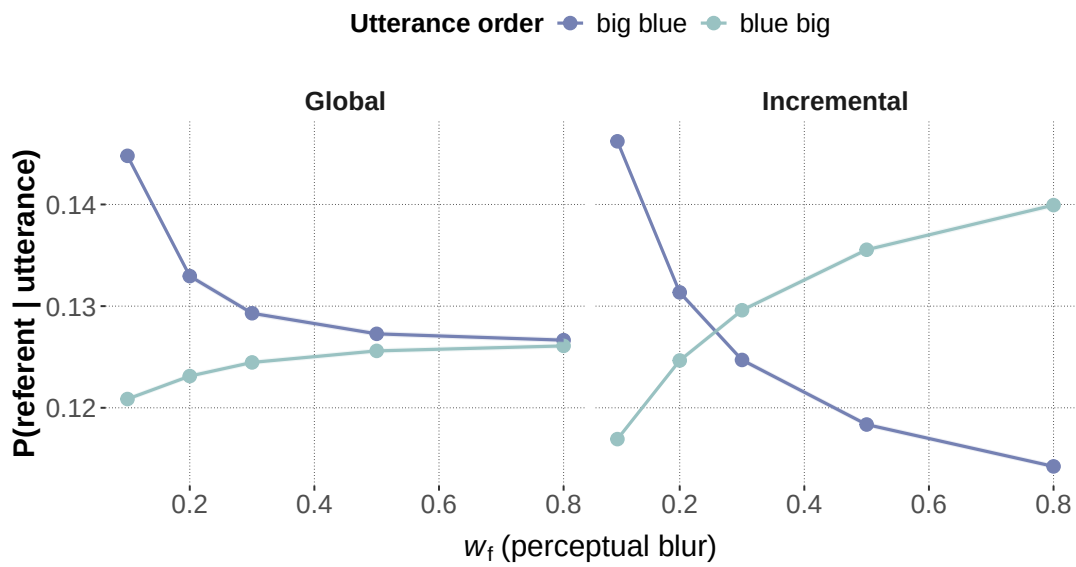


Figure E2

Effect of perceptual blur w_f on communicative success in the simulation.

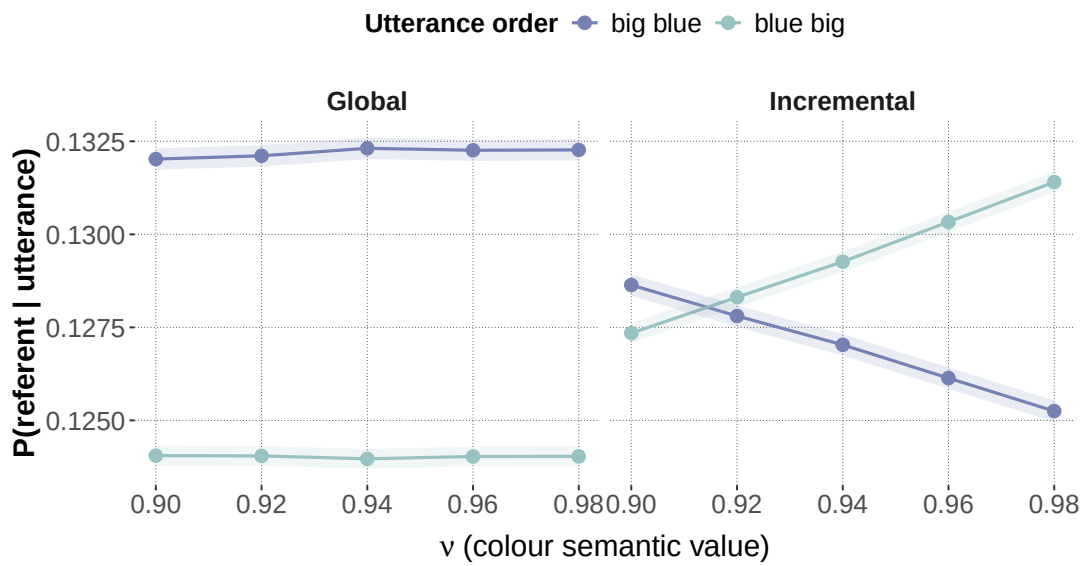


Figure E3

Effect of the colour/form semantic value v on communicative success in the simulation.

Appendix F

Slider Colour-Reliability Audit

The slider-rating model fixes the colour and form reliability ν at 0.80 (Section 6). Production uses the softer colour value $\nu_{\text{colour}} = 0.59$ in the final principled speaker (Section 6.2); for parity we apply the same kind of sensitivity audit to the slider’s incremental models. The audit samples $\nu \sim \text{Uniform}(0.5, 0.999)$ within the model and recomputes the literal listeners at each MCMC step, while keeping every other component of the incremental hierarchical speaker unchanged. Table F1 reports the resulting posteriors alongside the originally fixed- ν fits.

Table F1

Slider colour-reliability audit. Fixed- ν rows use $\nu = 0.80$; free- ν rows sample $\nu \sim \text{Uniform}(0.5, 0.999)$ and recompute literal listeners each step. “ELPD” is PSIS-LOO; ΔELPD and dse are relative to the incremental, context-updating fixed- ν model. All four models use 4 chains $\times$ 500 warmup $\times$ 500 samples; maximum split- $\hat{R}$ rounds to 1.01 for the free- ν models.

Model	ELPD	ΔELPD	dse	α (sd)	b (sd)	ν (94% HDI)
Incremental, updating, fixed $\nu = 0.80$	-2,774.1	0.0	—	0.69 (0.06)	0.58 (0.06)	—
Incremental, fixed, fixed $\nu = 0.80$	-2,774.1	-0.06	0.18	0.69 (0.06)	0.58 (0.06)	—
Incremental, updating, free ν	-2,775.0	-0.89	1.08	1.53 (0.61)	0.41 (0.15)	0.63 [0.50, 0.82]
Incremental, fixed, free ν	-2,774.3	-0.23	1.00	1.48 (0.55)	0.42 (0.13)	0.63 [0.51, 0.79]

Three points stand out. First, ν is weakly identified in the slider data: the posterior places substantial mass near the lower prior boundary at 0.5 and only just excludes the fixed value at the upper end, in contrast to the tight production posterior 0.59 [0.56, 0.62]. Second, the four models are statistically indistinguishable in PSIS-LOO ($|\Delta\text{ELPD}|/\text{dse} < 1$

throughout), so out-of-sample fit cannot decide between sharper and softer colour predicates in this response space. Third, under the freed posterior the rationality parameter α moves to roughly 1.5 (from 0.69) and the bias intercept b moves below 0.5 (from 0.58), because ν and the other parameters trade off: a softer colour predicate forces a stronger rationality term and reduces the residual size-first bias absorbed by b . We therefore retain the fixed $\nu = 0.80$ for the slider model in the main text and treat α and b at that specification as the reported quantities, while flagging the conditional status of the $b > 0.5$ finding (Section 6, Slider-Rating Study). The cross-dataset asymmetry — production requires a softer colour predicate in the final principled speaker, while the slider data do not strongly identify ν — is itself diagnostic of the response-space sensitivity argued for in the Cross-Dataset Summary.